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1. Application to a binary sampling problem. Sections ?? and ?? applied our bounds on $\mathcal{X} \subseteq \mathbb{R}^d$, where the intermediate distributions are log-concave and the Markov kernels are discretised diffusions. This section applies them on a finite space, $\mathcal{X} \subseteq \{0, 1\}^d$, where neither log-concavity nor smoothness is available, for the Bayesian variable selection problem.

Two pieces of the problem are already in the literature and we take them as given. Schäfer and Chopin (2013) showed that SMC is an effective sampler on large binary spaces, and their construction fixes the algorithmic choices we have to make: a tempering path, a Markov kernel, and an estimate of the normalising constant, which here is the marginal likelihood of the variable selection problem. Yang, Wainwright and Jordan (2016) analysed the natural local Metropolis–Hastings kernel on the same model and proved that its spectral gap at the posterior is at least of order $1/(ds_0^2)$, where s_0 is a truncation level on the model size; this is the only polynomial lower bound on a spectral gap available for this problem.

Neither reference supplies what the complexity bounds of Sections ??–?? need. Those bounds are driven by three quantities: the number T of intermediate distributions for which Assumption ?? holds; a lower bound on the spectral gap of the Markov kernel used at *each* of those distributions, not only at the last one, which is Assumption ??; and, for standard SMC, the β -mixing time τ_β rather than the spectral gap. Supplying the three is the object of this section, and the answers are Proposition 1.4, Theorems 1.24 and 1.32, and Lemma 1.33 respectively. They are combined in Theorem 1.34, which is the complexity statement, and Section 1.10 lists the algorithmic consequences.

The section is self-contained: every object is defined here, and the only external inputs are the four probabilistic events of Yang, Wainwright and Jordan (2016), isolated once and for all in Section 1.1.2, and Sinclair’s canonical path bound. We have kept the proofs long rather than short. Most of the difficulty is bookkeeping — which states a path visits, which pairs of states use a given edge, which of d and s_0 multiplies which constant — and bookkeeping errors are invisible in a compressed proof.

1.0.0.1. Road map. Section 1.1 sets up the model, the path, the kernel and the tools. Section 1.2 collects the deterministic inequalities that the good event of Yang, Wainwright and Jordan (2016) yields; they are used repeatedly afterwards and it is clearer to prove them once. Section 1.3 isolates the two conditions on the design and the one condition on the prior that everything else rests on, and shows what they cost. Section 1.4 bounds T . Sections 1.5 and 1.6 give two lower bounds on the spectral gap valid at every temperature: a crude one requiring nothing beyond that condition, and a sharp one requiring in addition that the temperature is not at a value where the mode of the intermediate target is changing. Section 1.7 computes the exact spectral gap on small instances and reports how much of these bounds is real. Section 1.8 converts spectral gaps into β -mixing times, Section 1.9 assembles the complexity, and Section 1.10 draws the practical conclusions.

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1.1. The model, the path and the sampler.

1.1.1. *The Bayesian variable selection model.* Let $y \in \mathbb{R}^n$ be a response vector and $X \in \mathbb{R}^{n \times d}$ a design matrix, whose columns $X_1, \dots, X_d$ are normalised so that $\|X_j\|_2^2 = n$. We assume the data are generated by the linear model

$$(1) \quad y = X\beta^* + w, \quad w \sim \mathcal{N}(0, \sigma_0^2 I_n),$$

with $\beta^* \in \mathbb{R}^d$ and $\sigma_0 > 0$ both unknown, and we work in the high-dimensional regime $d \geq n$.

A subset of the covariates is encoded by a binary vector $x \in \{0, 1\}^d$. We use the same letter for the binary vector and for the subset $\{j \in [d] : x_j = 1\}$ of $[d] := \{1, \dots, d\}$ that it indicates, so that $j \in x$, $x \cup \{j\}$, $x \subseteq x'$ and $|x| = \sum_j x_j$ all have their obvious meaning. We write X_x for the submatrix of X with columns indexed by x , and

$$P_x := X_x(X_x^\top X_x)^{-1}X_x^\top$$

for the orthogonal projection onto the span of those columns, with $P_\emptyset = 0$.

Fix a threshold $C_\beta > 0$. The set of *influential* covariates is

$$(2) \quad S := \{j \in [d] : |\beta_j^*| \geq C_\beta\}, \quad S^c := [d] \setminus S, \quad s^* := |S|,$$

and we write x^* for the binary vector indicating S . The coefficients $\beta_{S^c}^*$ off the support are allowed to be non-zero, provided they are small; this is Assumption A below. Throughout, the words *influential* and *null* refer to membership in S and in S^c , never to the value of β_j^* .

Following [Yang, Wainwright and Jordan \(2016\)](#), the Bayesian model puts a flat prior $p(\phi) \propto 1/\phi$ on the noise precision, Zellner's g -prior $\beta_x \mid x, \phi \sim \mathcal{N}(0, g\phi^{-1}(X_x^\top X_x)^{-1})$ on the regression coefficients, and the sparsity prior

$$(3) \quad p(x) = \frac{d^{-\kappa|x|}}{\sum_{z \in \mathcal{X}} d^{-\kappa|z|}} \quad \text{on} \quad \mathcal{X} := \{x \in \{0, 1\}^d : |x| \leq s_0\},$$

where $\kappa > 0$ and the truncation level s_0 satisfies $s^* < s_0 < n$. Integrating out (β_x, ϕ) against these priors gives the marginal likelihood

$$(4) \quad m(x) = (1 + g)^{-|x|/2} \{1 + g(1 - R_x^2)\}^{-n/2}, \quad R_x^2 := \frac{\|P_x y\|_2^2}{\|y\|_2^2},$$

up to a factor that does not depend on x and therefore cancels everywhere below; see [Yang, Wainwright and Jordan \(2016, Appendix A\)](#) for the computation. The posterior on models is $\pi(x) \propto p(x)m(x)$, and its normalising constant $\sum_{x \in \mathcal{X}} p(x)m(x)$ is the marginal likelihood of the variable selection problem, which is the quantity used for model comparison.

Two elementary properties of (4) are used constantly and it is worth recording them now. Both follow by inspection.

LEMMA 1.1. *For every $x \in \mathcal{X}$ and every $j \notin x$ with $|x| < s_0$,*

$$(5) \quad \frac{m(x \cup \{j\})}{m(x)} = (1 + g)^{-1/2} \left\{ 1 + \frac{R_{x \cup \{j\}}^2 - R_x^2}{g^{-1} + 1 - R_{x \cup \{j\}}^2} \right\}^{n/2},$$

and the numerator of the fraction satisfies

$$(6) \quad (R_{x \cup \{j\}}^2 - R_x^2) \|y\|_2^2 = \|(P_{x \cup \{j\}} - P_x)y\|_2^2,$$

while its denominator satisfies $(g^{-1} + 1 - R_{x \cup \{j\}}^2) \|y\|_2^2 = g^{-1} \|y\|_2^2 + \|(I - P_{x \cup \{j\}})y\|_2^2$.

PROOF. Write $z = x \cup \{j\}$. From (4),

$$\frac{m(z)}{m(x)} = (1+g)^{-1/2} \left\{ \frac{1+g(1-R_x^2)}{1+g(1-R_z^2)} \right\}^{n/2},$$

and dividing numerator and denominator of the inner fraction by g gives $\{g^{-1} + 1 - R_x^2\}/\{g^{-1} + 1 - R_z^2\}$, which equals $1 + (R_z^2 - R_x^2)/(g^{-1} + 1 - R_z^2)$. This is (5). For (6), note that $x \subset z$ implies $P_z P_x = P_x$, so $P_z - P_x$ is itself an orthogonal projection, whence $\|P_z y\|_2^2 - \|P_x y\|_2^2 = y^\top (P_z - P_x) y = \|(P_z - P_x) y\|_2^2$. Multiplying by $\|y\|_2^{-2}$ gives the claim. The last identity is $1 - R_z^2 = \|(I - P_z) y\|_2^2 / \|y\|_2^2$. $\square$

Formula (5) says that adding a covariate costs a fixed *Occam factor* $(1+g)^{-1/2}$ and buys a *likelihood gain* governed by $\|(P_{x \cup \{j\}} - P_x) y\|_2^2$, the amount of the response that the new column explains once the current ones have been projected out. Every computation below is a comparison of these two quantities.

1.1.2. Standing assumptions and the good event. We now restate the assumptions of Yang, Wainwright and Jordan (2016) in our notation. They are used only through the four events of Lemma 1.6, so a reader willing to take that lemma on trust may skip to Section 1.1.3.

ASSUMPTION 1.2 (Size of β^* , their Assumption A). There is a constant $\tilde{L} < \infty$ with

$$\|X \beta^*\|_2^2 \leq g \sigma_0^2 \log d, \quad \|X_{S^c} \beta_{S^c}^*\|_2^2 \leq \tilde{L} \sigma_0^2 \log d.$$

Two of the assumptions have to be imposed slightly above the truncation level, because Lemma 1.14 below compares a model $x \in \mathcal{X}$ with the model $x \cup x^*$, whose size may exceed s_0 . We therefore write

$$(7) \quad s_+ := s_0 + s^* \leq 2s_0$$

and impose Assumptions B and D of Yang, Wainwright and Jordan (2016) at level s_+ rather than s_0 . Since $s^* < s_0$, this is a strengthening by a factor at most two in the model size, and by nothing else.

ASSUMPTION 1.3 (Design, their Assumption B with $s = s_+$). There are constants $C_m > 0$ and $L < \infty$ such that, with $Z \sim \mathcal{N}(0, I_n)$,

$$\min_{|x| \leq s_+} \lambda_{\min}(n^{-1} X_x^\top X_x) \geq C_m, \quad \mathbb{E}_Z \left[\max_{|x| \leq s_+} \max_{k \in [d] \setminus x} n^{-1/2} |\langle (I - P_x) X_k, Z \rangle| \right] \leq \frac{1}{2} \sqrt{L C_m \log d}.$$

ASSUMPTION 1.4 (Hyperparameters, their Assumption C). There are $\alpha > 0$ and $1 \leq c_1 \leq c_2 < \infty$ with $c_1 d^{2\alpha} \leq g \leq c_2 d^{2\alpha}$, and $\kappa + \alpha \geq c_\alpha(L + \tilde{L}) + 2$ for the universal constant c_α of Yang, Wainwright and Jordan (2016).

ASSUMPTION 1.5 (Sparsity control, their Assumption D at level s_+). With $\omega(X) := \max_{|x| \leq s_+} \|(X_x^\top X_x)^{-1} X_x^\top X_{S \setminus x}\|_{\text{op}}^2$ and constants $r > 4$ and $K \geq 4 + \alpha + c\tilde{L}$,

$$(2C_m^{-2} \omega(X) + 1) s^* \leq s_0, \quad s_+ \leq \frac{1}{8rK} \left\{ \frac{n}{\log d} - 16\tilde{L} \sigma_0^2 \right\}.$$

The constant r here is the constant C_0 of Yang, Wainwright and Jordan (2016); we use the same letter for it and for the constant appearing in the event B_n of Lemma 1.6, which is

legitimate because both statements are monotone in that constant and we may take the larger of the two.

We assume Assumptions 1.2–1.5 throughout, together with the sample size requirement

$$(8) \quad n \geq 32(r s_+ + \tilde{L}) \log d,$$

which is Assumption 1.5 with a different numerical constant and is used only in Lemma 1.14.

LEMMA 1.6 (Good event). *Let $\mathcal{X}^+ := \{x \in \{0, 1\}^d : |x| \leq s_+\}$ and define*

$$(9) \quad \begin{aligned} A_n &:= \left\{ \max_{x_2 \subset x_1 \in \mathcal{X}^+} \frac{w^\top (P_{x_1} - P_{x_2}) w}{|x_1| - |x_2|} \leq L \sigma_0^2 \log d \right\}, & B_n &:= \left\{ \max_{x \in \mathcal{X}^+} \frac{w^\top P_x w}{|x|} \leq r \sigma_0^2 \log d \right\}, \\ C_n &:= \left\{ \left| \frac{\|w\|_2^2}{n \sigma_0^2} - 1 \right| \leq \frac{1}{2} \right\}, & D_n &:= \left\{ \frac{\|y\|_2^2}{g} \leq \frac{5 n \sigma_0^2}{s^*} \right\}, \end{aligned}$$

and $\mathcal{H}_n := A_n \cap B_n \cap C_n \cap D_n$. Then $\Pr(\mathcal{H}_n) \geq 1 - 6d^{-c}$ for a universal constant $c > 0$.

PROOF. This is Lemma 3 of Yang, Wainwright and Jordan (2016) applied with their truncation level set to s_+ instead of s_0 . Their model class is $\{x : |x| \leq s_0\}$ and enters their proof only through that bound; the hypotheses of their lemma are Assumptions A, B(s_0), C and D(s_0), which for the truncation level s_+ read exactly our Assumptions 1.2, 1.3, 1.4 and 1.5. With that substitution their model class is $\mathcal{X}^+$ and their four events are (9).

It is worth recording why the substitution is harmless, since the events are maxima over exponentially many models. For B_n , the variable $w^\top P_x w / \sigma_0^2$ is χ^2 with $|x|$ degrees of freedom, and there are $\binom{d}{\ell} \leq d^\ell$ models of size ℓ , so a union bound and a standard χ^2 tail give

$$\Pr(B_n^c) \leq \sum_{\ell=1}^{s_+} \binom{d}{\ell} \Pr(\chi_\ell^2 \geq r \ell \log d) \leq \sum_{\ell \geq 1} d^\ell e^{-(r/4)\ell \log d} \leq 2d^{-(r/4-1)},$$

which does not depend on the truncation level at all: the factor ℓ in the exponent absorbs the d^ℓ from the union bound, and the series converges because $r > 4$. The same mechanism governs A_n , where the degrees of freedom are $|x_1| - |x_2|$ and the counting is controlled by the sparse projection condition of Assumption 1.3; C_n does not involve the model class; and D_n involves it only through Assumption 1.5. $\square$

From now on all statements that mention $\mathcal{H}_n$ are deterministic statements about a fixed realisation of (y, w) lying in that event. No further probabilistic argument occurs in this section.

1.1.3. *The tempering path.* We take the sparsity prior as the dominating measure ν of Section ?? and temper the marginal likelihood only:

$$(10) \quad \nu := p, \quad g_t := m^{\lambda_t}, \quad \pi_t(dx) = \frac{m(x)^{\lambda_t} p(dx)}{Z(\lambda_t)}, \quad Z(\lambda) := p(m^\lambda) = \sum_{x \in \mathcal{X}} p(x) m(x)^\lambda,$$

along a schedule $0 = \lambda_{-1} < \lambda_0 < \dots < \lambda_T = 1$. Then $G_t = m^{\lambda_t - \lambda_{t-1}}$, and $\pi_{-1} = \nu = p$ with $Z_{-1} = Z(0) = 1$ at one end, $\pi_T = \pi$ with $Z_T = p(m)$ at the other; the SMC estimate $\hat{Z}_T$ therefore targets the marginal likelihood directly. For an arbitrary $\lambda \geq 0$ we write

$$(11) \quad \pi_\lambda(x) := \frac{m(x)^\lambda p(x)}{Z(\lambda)}, \quad \text{so that} \quad \pi_t = \pi_{\lambda_t}, \quad \pi_0 = p, \quad \pi_1 = \pi.$$

Since $\mathcal{X}$ is finite and $m > 0$ on it, $Z(\lambda) \in (0, \infty)$ and π_λ is well defined for every real λ ; we shall use $\lambda \in [0, 2]$.

Three properties of the path (10) are used repeatedly.

- (P1) *ν can be sampled exactly.* Draw $k \in \{0, \dots, s_0\}$ from the law proportional to $\binom{d}{k} d^{-\kappa k}$, then draw a uniform k -subset of $[d]$. Indeed $p(x)$ depends on x only through $|x|$, and there are $\binom{d}{k}$ vectors with $|x| = k$.
- (P2) *The prior does not distort levels.* Because p depends on x only through $|x|$, the restriction of π_λ to a level set $\{|x| = k\}$ is proportional to m^λ there; at $\lambda = 0$ it is uniform on that level.
- (P3) *The penalty is temperature-free.* For any x and $j \notin x$ with $|x| < s_0$,

$$(12) \quad \frac{\pi_\lambda(x \cup \{j\})}{\pi_\lambda(x)} = d^{-\kappa} \left\{ \frac{m(x \cup \{j\})}{m(x)} \right\}^\lambda,$$

so the factor $d^{-\kappa}$ charged for each extra covariate is the same at every temperature, whereas the likelihood gain is damped by λ .

1.1.4. *The Markov kernel.* At temperature λ we use the lazy version of the local Metropolis–Hastings kernel of Yang, Wainwright and Jordan (2016), which we call the *flip-and-exchange* kernel and denote by K_λ . From the current state $x \in \mathcal{X}$ it acts as follows.

- (K1) With probability $1/2$, stay at x .
- (K2) Otherwise, with probability $1/2$, draw $j \in [d]$ uniformly and propose x' with $x'_j = 1 - x_j$ (a *flip*).
- (K3) Otherwise, draw (k, ℓ) uniformly in $\{j : x_j = 1\} \times \{j : x_j = 0\}$ and propose x' with $x'_k = 0, x'_\ell = 1$ (an *exchange*); if $x = \emptyset$, stay at x .
- (K4) Accept a proposal x' with probability $\min\{1, \pi_\lambda(x')/\pi_\lambda(x)\}$ if $x' \in \mathcal{X}$, and reject it otherwise.

One step costs one evaluation of m , that is $\mathcal{O}(ns_0 + s_0^2)$ operations if the Cholesky factor of $X_x^\top X_x$ is updated by rank one. The properties of K_λ that the rest of the section uses are the following.

LEMMA 1.7. *For every $\lambda \geq 0$:*

- (i) $\pi_\lambda(x) > 0$ for every $x \in \mathcal{X}$;
- (ii) K_λ is π_λ -reversible;
- (iii) K_λ is irreducible and aperiodic on $\mathcal{X}$;
- (iv) K_λ is non-negative definite as an operator on $\mathcal{L}_{\pi_\lambda}^2$.

PROOF. i By (4), $m(x) \in (0, \infty)$ for every x , since $R_x^2 \in [0, 1]$ and $g > 0$ make both factors finite and positive; and $p(x) > 0$ by (3). Hence $Z(\lambda) \in (0, \infty)$ and $\pi_\lambda(x) > 0$.

ii Write $q(x, x')$ for the probability that the kernel proposes x' from x . Within scheme (K2), $q(x, x') = 1/(4d)$ whenever x and x' differ by one coordinate, which is symmetric in (x, x') . Within scheme (K3), if x' is obtained from x by moving a coordinate from 1 to 0 and another from 0 to 1, then $|x'| = |x|$ and $q(x, x') = (4|x|(d - |x|))^{-1} = q(x', x)$. So q is symmetric, and for $x \neq x'$ in $\mathcal{X}$,

$$\pi_\lambda(x) K_\lambda(x, x') = \pi_\lambda(x) q(x, x') \min\left\{1, \frac{\pi_\lambda(x')}{\pi_\lambda(x)}\right\} = q(x, x') \min\{\pi_\lambda(x), \pi_\lambda(x')\},$$

which is symmetric in (x, x') . This is detailed balance, and it is also the identity used throughout in Lemma 1.8.

iii Let $x \neq x'$ in $\mathcal{X}$. Delete the elements of x one at a time and then add those of x' one at a time; every intermediate state is a subset of x or of x' , hence lies in $\mathcal{X}$, and consecutive states differ by one flip, so each transition has positive probability by i and (13). Hence K_λ is irreducible. Aperiodicity is $K_\lambda(x, x) \geq 1/2 > 0$, which holds by (K1).

iv Let $\tilde{K}_\lambda$ be the kernel obtained by deleting (K1), so that $K_\lambda = \frac{1}{2}I + \frac{1}{2}\tilde{K}_\lambda$. The computation in ii shows that $\tilde{K}_\lambda$ is also π_λ -reversible, hence self-adjoint on $\mathcal{L}_{\pi_\lambda}^2$, and it is a Markov kernel, hence a contraction there. Its spectrum therefore lies in $[-1, 1]$ and that of K_λ in $[0, 1]$. $\square$

The single quantitative consequence of (K1)–(K4) that we use is the following lower bound on the edge weights of the chain. Recall that for a μ -reversible kernel K we write $Q(x, x') := \mu(x)K(x, x')$; it is symmetric in (x, x') .

LEMMA 1.8 (Edge weights). *Let $\lambda \geq 0$ and let $x \neq x'$ be two states of $\mathcal{X}$. Write $Q_\lambda(x, x') = \pi_\lambda(x)K_\lambda(x, x')$. If x and x' differ by a single flip, then*

$$(13) \quad Q_\lambda(x, x') \geq \frac{1}{4d} \min\{\pi_\lambda(x), \pi_\lambda(x')\},$$

and if they differ by an exchange, then $Q_\lambda(x, x') \geq (4ds_0)^{-1} \min\{\pi_\lambda(x), \pi_\lambda(x')\}$.

PROOF. Suppose first that x' is obtained from x by flipping coordinate j . The probability that the kernel proposes x' is $\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{d} = 1/(4d)$ by (K1) and (K2). Since $x' \in \mathcal{X}$ by hypothesis, the proposal is accepted with probability $\min\{1, \pi_\lambda(x')/\pi_\lambda(x)\}$, whence

$$\pi_\lambda(x)K_\lambda(x, x') \geq \frac{\pi_\lambda(x)}{4d} \min\left\{1, \frac{\pi_\lambda(x')}{\pi_\lambda(x)}\right\} = \frac{1}{4d} \min\{\pi_\lambda(x), \pi_\lambda(x')\}.$$

If instead x' is obtained from x by exchanging $k \in x$ and $\ell \notin x$, the probability of proposing x' is $\frac{1}{2} \cdot \frac{1}{2} \cdot (|x|(d - |x|))^{-1} \geq (4ds_0)^{-1}$, because $|x| \leq s_0$ and $d - |x| \leq d$. The same computation applies. $\square$

The two constants $1/(4d)$ and $1/(4ds_0)$ differ by the factor s_0 , and keeping track of which of the two a proof uses is what separates Theorem 1.31 from Theorem 2 of Yang, Wainwright and Jordan (2016); see Remark 1.7.

1.1.5. *Dirichlet forms, decomposition and canonical paths.* We collect here the three tools used in Sections 1.5 and 1.6. Throughout, μ is a probability measure with full support on a finite set $\mathcal{X}$ and K is a μ -reversible Markov kernel on $\mathcal{X}$.

1.1.5.1. *Dirichlet form and spectral gap.* The Dirichlet form of (K, μ) is

$$(14) \quad \mathcal{E}_{K, \mu}(f) := \frac{1}{2} \sum_{x, y \in \mathcal{X}} \mu(x)K(x, y)\{f(x) - f(y)\}^2 = \langle f, (I - K)f \rangle_\mu,$$

and the *Poincaré constant* of (K, μ) is the largest $\gamma \geq 0$ such that $\gamma \text{Var}_\mu(f) \leq \mathcal{E}_{K, \mu}(f)$ for all $f : \mathcal{X} \rightarrow \mathbb{R}$.

It equals $1 - \lambda_2(K)$, where $1 = \lambda_1(K) > \lambda_2(K) \geq \dots$ are the eigenvalues of K as a self-adjoint operator on $\mathcal{L}_\mu^2$, provided K is irreducible. Indeed, K is self-adjoint by reversibility, $K\mathbf{1} = \mathbf{1}$, and $\mathbf{1}$ spans the eigenspace of λ_1 when K is irreducible, so $\mathcal{L}_{0, \mu}^2$ is K -invariant and $\lambda_2 = \max\{\langle f, Kf \rangle_\mu : f \in \mathcal{L}_{0, \mu}^2, \|f\|_\mu = 1\}$ by the variational characterisation of eigenvalues. For $f \in \mathcal{L}_{0, \mu}^2$ one has $\text{Var}_\mu(f) = \|f\|_\mu^2$ and, by (14), $\mathcal{E}_{K, \mu}(f) = \|f\|_\mu^2 - \langle f, Kf \rangle_\mu$; taking

the infimum of $\mathcal{E}_{K,\mu}(f)/\text{Var}_\mu(f)$ over that subspace gives $1 - \lambda_2$, and adding a constant to f changes neither side.

If moreover K is non-negative definite, so that all its eigenvalues lie in $[0, 1]$, then $\sup\{\|Kh\|_\mu : h \in \mathcal{L}_{0,\mu}^2, \|h\|_\mu = 1\} = \lambda_2$, so the Poincaré constant coincides with the absolute spectral gap of Section ??; we then write $\text{Gap}(K)$ for it without ambiguity. By Lemma 1.7 the kernel K_λ qualifies. This is the only role of (K1), and it matters, because Assumption ?? and Lemma 1.33 both refer to the absolute gap, which for a reversible kernel that is not non-negative definite may be much smaller than $1 - \lambda_2$.

1.1.5.2. Restriction.. The next lemma records the elementary properties of the chain obtained by forbidding the moves that leave a block. It is used three times below and it is convenient to have it once.

LEMMA 1.9 (Restriction). *Let $\mathcal{X} = \bigsqcup_{a \in \mathcal{A}} B_a$ be a partition into non-empty blocks, and suppose $K(x, x) \geq \frac{1}{2}$ for every x . For $a \in \mathcal{A}$ set $\mu_a := \mu(\cdot \mid B_a)$, $\bar{\mu}(a) := \mu(B_a)$, and define K_a on B_a by*

$$(15) \quad K_a(x, y) := K(x, y) \quad (x \neq y \in B_a), \quad K_a(x, x) := 1 - \sum_{y \in B_a \setminus \{x\}} K(x, y).$$

Then, for each a , K_a is a Markov kernel on B_a , it is μ_a -reversible, it satisfies $K_a(x, x) \geq \frac{1}{2}$ and it is non-negative definite on $\mathcal{L}_{\mu_a}^2$. Moreover

$$(16) \quad \sum_{a \in \mathcal{A}} \bar{\mu}(a) \mathcal{E}_{K_a, \mu_a}(f) \leq \mathcal{E}_{K, \mu}(f) \quad \text{for every } f : \mathcal{X} \rightarrow \mathbb{R},$$

so that if the Poincaré constant of (K_a, μ_a) is at least γ_{in} for every a , then the first inequality of (17) holds with that γ_{in} .

PROOF. Fix a . The off-diagonal entries of K_a are non-negative and its rows sum to 1 by construction, and

$$K_a(x, x) = 1 - \sum_{y \in B_a \setminus \{x\}} K(x, y) \geq 1 - \sum_{y \neq x} K(x, y) = K(x, x) \geq \frac{1}{2},$$

so K_a is a Markov kernel with the stated laziness. For reversibility, if $x \neq y$ in B_a then $\bar{\mu}(a)\mu_a(x)K_a(x, y) = \mu(x)K(x, y)$, which is symmetric in (x, y) because K is μ -reversible; dividing by $\bar{\mu}(a) > 0$ gives $\mu_a(x)K_a(x, y) = \mu_a(y)K_a(y, x)$, and the case $x = y$ is trivial. For non-negative definiteness, put $R := 2K_a - I$. Its off-diagonal entries are $2K_a(x, y) \geq 0$, its diagonal entries are $2K_a(x, x) - 1 \geq 0$, and its rows sum to $2 - 1 = 1$, so R is a Markov kernel; it is μ_a -reversible because K_a is, hence self-adjoint on $\mathcal{L}_{\mu_a}^2$ with spectrum in $[-1, 1]$. Therefore $K_a = \frac{1}{2}(I + R)$ has spectrum in $[0, 1]$.

For (16), the diagonal terms of the Dirichlet form vanish, so

$$\sum_a \bar{\mu}(a) \mathcal{E}_{K_a, \mu_a}(f) = \frac{1}{2} \sum_a \sum_{x \neq y \in B_a} \mu(x) K(x, y) \{f(x) - f(y)\}^2 \leq \frac{1}{2} \sum_{x, y \in \mathcal{X}} \mu(x) K(x, y) \{f(x) - f(y)\}^2 = \mathcal{E}_{K, \mu}(f),$$

because the ordered pairs lying in a common block form a subset of all ordered pairs and every summand is non-negative. The last claim follows by applying $\text{Var}_{\mu_a}(f) \leq \gamma_{\text{in}}^{-1} \mathcal{E}_{K_a, \mu_a}(f)$ for each a and averaging against $\bar{\mu}$. $\square$

1.1.5.3. Decomposition.. The next lemma reduces a Poincaré inequality on $\mathcal{X}$ to two Poincaré inequalities, one inside the blocks of a partition and one on the induced chain between blocks. It is a variant of the decomposition inequalities of [Madras and Randall \(2002\)](#) and [Jerrum et al. \(2004\)](#); we give the proof in full because the constants matter.

LEMMA 1.10 (Decomposition). *Let $\mathcal{X} = \bigsqcup_{a \in \mathcal{A}} B_a$ be a partition into non-empty blocks, write $a(x)$ for the index with $x \in B_{a(x)}$, and set*

$$\mu_a := \mu(\cdot \mid B_a), \quad \bar{\mu}(a) := \mu(B_a), \quad \bar{f}(a) := \mathbb{E}_{\mu_a}[f],$$

and let $\bar{K}$ be the kernel on $\mathcal{A}$ defined for $a \neq b$ by $\bar{K}(a, b) := \bar{\mu}(a)^{-1} \sum_{x \in B_a} \mu(x) K(x, B_b)$, with the diagonal fixed by $\sum_b \bar{K}(a, b) = 1$. Suppose that there are $\gamma_{\text{in}}, \bar{\gamma} \in (0, 1]$ such that

$$(17) \quad \sum_{a \in \mathcal{A}} \bar{\mu}(a) \text{Var}_{\mu_a}(f) \leq \frac{1}{\gamma_{\text{in}}} \mathcal{E}_{K, \mu}(f) \quad \text{for all } f, \quad \text{Var}_{\bar{\mu}}(\bar{g}) \leq \frac{1}{\bar{\gamma}} \mathcal{E}_{\bar{K}, \bar{\mu}}(\bar{g}) \quad \text{for all } \bar{g}.$$

Then $\text{Var}_{\mu}(f) \leq 7(\bar{\gamma}\gamma_{\text{in}})^{-1} \mathcal{E}_{K, \mu}(f)$ for all f . In particular, if K is non-negative definite then $\text{Gap}(K) \geq \gamma_{\text{in}}\bar{\gamma}/7$.

PROOF. Write $Q(x, y) = \mu(x)K(x, y)$ and, for $a \neq b$, $Q_{ab} := \sum_{x \in B_a} \sum_{y \in B_b} Q(x, y)$, and set

$$V_{\text{in}} := \sum_a \bar{\mu}(a) \text{Var}_{\mu_a}(f) = \sum_{x \in \mathcal{X}} \mu(x) \{f(x) - \bar{f}(a(x))\}^2,$$

the second equality being the definition of Var_{μ_a} written out.

Step 1: $\bar{K}$ is $\bar{\mu}$ -reversible. For $a \neq b$, $\bar{\mu}(a)\bar{K}(a, b) = \sum_{x \in B_a} \mu(x)K(x, B_b) = Q_{ab}$, and $Q_{ab} = Q_{ba}$ because Q is symmetric. Hence $\mathcal{E}_{\bar{K}, \bar{\mu}}(\bar{g}) = \frac{1}{2} \sum_{a \neq b} Q_{ab} \{\bar{g}(a) - \bar{g}(b)\}^2$, and the diagonal of $\bar{K}$ plays no role.

Step 2: the projected Dirichlet form is controlled. Fix $a \neq b$, $x \in B_a$ and $y \in B_b$. Decomposing

$$\bar{f}(a) - \bar{f}(b) = \{\bar{f}(a) - f(x)\} + \{f(x) - f(y)\} + \{f(y) - \bar{f}(b)\}$$

and using $(u + v + w)^2 \leq 3(u^2 + v^2 + w^2)$ gives

$$\{\bar{f}(a) - \bar{f}(b)\}^2 \leq 3[\{f(x) - \bar{f}(a)\}^2 + \{f(x) - f(y)\}^2 + \{f(y) - \bar{f}(b)\}^2].$$

Multiply by $Q(x, y) \geq 0$ and sum over $x \in B_a$ and $y \in B_b$; the left-hand side becomes $Q_{ab}\{\bar{f}(a) - \bar{f}(b)\}^2$. Now sum over ordered pairs $a \neq b$ and multiply by $\frac{1}{2}$. The left-hand side becomes $\mathcal{E}_{\bar{K}, \bar{\mu}}(\bar{f})$ by Step 1. On the right, the middle term contributes

$$\frac{3}{2} \sum_{a \neq b} \sum_{x \in B_a, y \in B_b} Q(x, y) \{f(x) - f(y)\}^2 \leq \frac{3}{2} \sum_{x, y \in \mathcal{X}} Q(x, y) \{f(x) - f(y)\}^2 = 3\mathcal{E}_{K, \mu}(f),$$

since the pairs (x, y) lying in distinct blocks form a subset of all pairs. The first term contributes

$$\frac{3}{2} \sum_{a \neq b} \sum_{x \in B_a, y \in B_b} Q(x, y) \{f(x) - \bar{f}(a)\}^2 = \frac{3}{2} \sum_{x \in \mathcal{X}} \mu(x) K(x, \mathcal{X} \setminus B_{a(x)}) \{f(x) - \bar{f}(a(x))\}^2 \leq \frac{3}{2} V_{\text{in}},$$

where we first summed over y and then over $b \neq a$, and used $K(x, \mathcal{X} \setminus B_{a(x)}) \leq 1$. The third term is equal to the first after exchanging the roles of (a, x) and (b, y) , which is legitimate because we are summing over ordered pairs. Altogether

$$(18) \quad \mathcal{E}_{\bar{K}, \bar{\mu}}(\bar{f}) \leq 3\mathcal{E}_{K, \mu}(f) + 3V_{\text{in}}.$$

Step 3: conclusion. By the law of total variance, $\text{Var}_\mu(f) = V_{\text{in}} + \text{Var}_{\bar{\mu}}(\bar{f})$. Applying the second inequality of (17) to $\bar{g} = \bar{f}$ and then (18),

$$\text{Var}_\mu(f) \leq V_{\text{in}} + \frac{3\mathcal{E}_{K,\mu}(f) + 3V_{\text{in}}}{\bar{\gamma}} \leq \left(\frac{1}{\gamma_{\text{in}}} + \frac{3}{\bar{\gamma}} + \frac{3}{\bar{\gamma}\gamma_{\text{in}}} \right) \mathcal{E}_{K,\mu}(f),$$

using the first inequality of (17) twice. Since $\gamma_{\text{in}}, \bar{\gamma} \leq 1$ we have $1/\gamma_{\text{in}} \leq 1/(\bar{\gamma}\gamma_{\text{in}})$ and $3/\bar{\gamma} \leq 3/(\bar{\gamma}\gamma_{\text{in}})$, so the bracket is at most $7/(\bar{\gamma}\gamma_{\text{in}})$. $\square$

REMARK 1.1. *The first inequality in (17) is not the statement that each restricted chain has a Poincaré constant γ_{in} ; it is that statement summed against $\bar{\mu}$, which is weaker and is what Proposition 1.5 proves. The distinction matters because the restricted chains are indexed by 2^{s^*} blocks and we do not want to lose a union bound over them.*

1.1.5.4. Canonical paths. A canonical path ensemble Γ for (K, μ) assigns to each ordered pair (x, x') of distinct states a simple path $\Gamma_{x,x'} = (x = z_0, z_1, \dots, z_k = x')$ in the graph on $\mathcal{X}$ whose edges are the pairs $e = (z, z')$ with $Q(z, z') > 0$. Its length and congestion are

$$(19) \quad \ell(\Gamma) := \max_{x \neq x'} |\Gamma_{x,x'}|, \quad \varrho(\Gamma) := \max_e \frac{1}{Q(e)} \sum_{(x,x'): e \in \Gamma_{x,x'}} \mu(x)\mu(x'),$$

the maximum being over directed edges. Sinclair (1992) proved that

$$(20) \quad \gamma \geq \frac{1}{\varrho(\Gamma) \ell(\Gamma)}$$

for the Poincaré constant γ of (K, μ) .

All our ensembles are built by the same device, which we state once.

DEFINITION 1.11 (Ensemble generated by a map). Let $F : \mathcal{X} \rightarrow \mathcal{X}$ and let $\Phi : \mathcal{X} \rightarrow \mathbb{N}$ satisfy $\Phi(F(z)) = \Phi(z) - 1$ whenever $\Phi(z) \geq 1$, and $\{\Phi = 0\} = \{z_\bullet\}$ for a single state $z_\bullet$ with $F(z_\bullet) = z_\bullet$. The orbit of x is $\mathcal{O}(x) := (x, F(x), \dots, F^{\Phi(x)}(x) = z_\bullet)$. The ensemble generated by F assigns to (x, x') the path that follows $\mathcal{O}(x)$ up to the first state v that also lies on $\mathcal{O}(x')$, and then follows $\mathcal{O}(x')$ backwards from v to x' .

LEMMA 1.12. *In the setting of Definition 1.11, suppose in addition that $Q(z, F(z)) > 0$ for every $z \neq z_\bullet$. Then the ensemble generated by F is a canonical path ensemble, with $\ell(\Gamma) \leq 2 \max_x \Phi(x)$; in particular K is irreducible, since any two states are joined by a path of edges of positive weight. Moreover, writing*

$$(21) \quad \text{Prec}(z) := \{x \in \mathcal{X} : z \in \mathcal{O}(x)\}$$

for the set of precedents of z , every directed edge $e = (z, F(z))$ satisfies

$$(22) \quad \sum_{(x,x'): e \in \Gamma_{x,x'}} \mu(x)\mu(x') \leq \mu[\text{Prec}(z)].$$

PROOF. *Orbits are simple paths.* Since $\Phi(F(z)) = \Phi(z) - 1$ whenever $\Phi(z) \geq 1$, the values $\Phi(x), \Phi(F(x)), \dots$ strictly decrease until they reach 0, so $F^{\Phi(x)}(x)$ is the first iterate with $\Phi = 0$, that is $z_\bullet$; the $\Phi(x) + 1$ states of $\mathcal{O}(x)$ carry distinct values of Φ and are therefore distinct. Each consecutive pair is an edge by the hypothesis $Q(z, F(z)) > 0$.

The meeting point. Both orbits contain $z_\bullet$, so $\mathcal{O}(x) \cap \mathcal{O}(x') \neq \emptyset$ and v , the first state of $\mathcal{O}(x)$ lying on $\mathcal{O}(x')$, is well defined.

The concatenation is a simple path. Write $\mathcal{O}(x) = (x, \dots, v, \dots, z_\bullet)$ and $\mathcal{O}(x') = (x', \dots, v, \dots, z_\bullet)$, and let P and P' be the portions strictly before v of the first and of the second. No state of P lies on $\mathcal{O}(x')$, by the minimality of v in $\mathcal{O}(x)$; in particular $P \cap P' = \emptyset$ and $v \notin P$, $v \notin P'$. Hence the states of P , the state v , and the states of P' are pairwise distinct, and the concatenation $P \rightarrow v \rightarrow \overline{P'}$ is a simple path from x to x' . Its length is $\{\Phi(x) - \Phi(v)\} + \{\Phi(x') - \Phi(v)\} \leq \Phi(x) + \Phi(x') \leq 2 \max_z \Phi(z)$.

Proof of (22). Fix the directed edge $e = (z, F(z))$ and a pair (x, x') whose path contains it. The path consists of the edges $(w, F(w))$ for $w \in P$ followed by the edges $(F(w), w)$ for $w \in P'$. If e were one of the latter we would have $z = F(w)$ and $F(z) = w$ for some w , hence $F^2(z) = z$ and $\Phi(z) = \Phi(z) - 2$, which is impossible. So $e = (w, F(w))$ for some $w \in P \subseteq \mathcal{O}(x)$, whence $z = w \in \mathcal{O}(x)$, that is $x \in \text{Prec}(z)$. Therefore

$$\sum_{(x, x') : e \in \Gamma_{x, x'}} \mu(x) \mu(x') \leq \sum_{x \in \text{Prec}(z)} \sum_{x' \in \mathcal{X}} \mu(x) \mu(x') = \mu[\text{Prec}(z)].$$

The reverse directed edge $(F(z), z)$ is used only by pairs with $x' \in \text{Prec}(z)$ and obeys the same bound, and Q gives it the same weight, so the maximum in (19) may be restricted to edges of the form $(z, F(z))$. $\square$

Lemma 1.12 reduces every congestion computation below to two questions: how large is $\mu[\text{Prec}(z)]/\mu(z)$, and how small can $Q(z, F(z))/\mu(z)$ be. The second is answered by Lemma 1.8 as soon as we know that F does not move downhill too fast.

1.2. *Deterministic consequences of the good event.* Every statement in the remainder of this section that refers to the data goes through the three inequalities of this subsection. They are stated for a fixed realisation lying in $\mathcal{H}_n$ and are purely deterministic. We use the abbreviations

$$(23) \quad E := \frac{\|X\beta^*\|_2^2}{\sigma_0^2} \quad (\text{the signal energy}), \quad c_g := \frac{1}{2} \log(2c_2), \quad s_+^* := \max(s^*, 2).$$

Note that E is a signal-to-noise ratio and not a variance; it is bounded by $g \log d$ by Assumption 1.2, and it will be of order $s^* \log d$ in the regime of interest (Remark 1.5).

LEMMA 1.13 (Occam factor). *Under Assumption 1.4, $\alpha \log d \leq \frac{1}{2} \log(1 + g) \leq \alpha \log d + c_g$.*

PROOF. For the upper bound, $g \geq c_1 \geq 1$ gives $1 + g \leq 2g \leq 2c_2 d^{2\alpha}$, so $\frac{1}{2} \log(1 + g) \leq \frac{1}{2} \log(2c_2) + \alpha \log d$. For the lower bound, $1 + g \geq g \geq c_1 d^{2\alpha} \geq d^{2\alpha}$. $\square$

LEMMA 1.14 (Uniform residual lower bound). *On $\mathcal{H}_n$, and under (8), every $z \in \mathcal{X}$ satisfies*

$$(24) \quad \|(I - P_z)y\|_2^2 \geq \frac{n\sigma_0^2}{8}.$$

PROOF. Fix $z \in \mathcal{X}$ and set $z^+ := z \cup x^*$, so that $|z^+| \leq |z| + s^* \leq s_+$ and $z^+ \in \mathcal{X}^+$. Since $z \subseteq z^+$, the range of P_z is contained in that of P_{z^+} , hence $I - P_z \succcurlyeq I - P_{z^+}$ and

$$(25) \quad \|(I - P_z)y\|_2^2 \geq \|(I - P_{z^+})y\|_2^2.$$

Because $S \subseteq z^+$, the vector $X_S \beta_S^*$ lies in the range of P_{z^+} , so $(I - P_{z^+})X_S \beta_S^* = 0$ and

$$(I - P_{z^+})y = (I - P_{z^+})w + (I - P_{z^+})X_{S^c} \beta_{S^c}^*.$$

Apply the elementary inequality $\|a + b\|_2^2 \geq \frac{1}{2}\|a\|_2^2 - \|b\|_2^2$, which is $\frac{1}{2}(\|a\|_2 - 2\|b\|_2)^2 \geq 0$ rearranged, with $a = (I - P_{z^+})w$ and $b = (I - P_{z^+})X_{S^c}\beta_{S^c}^*$. Since projections are non-expansive, $\|b\|_2^2 \leq \|X_{S^c}\beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log d$ by Assumption 1.2, so

$$\|(I - P_{z^+})y\|_2^2 \geq \frac{1}{2}(\|w\|_2^2 - w^\top P_{z^+} w) - \tilde{L}\sigma_0^2 \log d.$$

On C_n we have $\|w\|_2^2 \geq n\sigma_0^2/2$, and on B_n , applied to $z^+ \in \mathcal{X}^+$, we have $w^\top P_{z^+} w \leq r s_+ \sigma_0^2 \log d$. Therefore

$$\|(I - P_{z^+})y\|_2^2 \geq \frac{n\sigma_0^2}{4} - \left\{ \frac{r s_+}{2} + \tilde{L} \right\} \sigma_0^2 \log d \geq \frac{n\sigma_0^2}{4} - \{r s_+ + \tilde{L}\} \sigma_0^2 \log d.$$

By (8) the subtracted term is at most $n\sigma_0^2/32$, so the right-hand side is at least $7n\sigma_0^2/32 \geq n\sigma_0^2/8$. Combining with (25) finishes the proof. $\square$

Inequality (24) is the workhorse. It says that no model in $\mathcal{X}$, however large or however well chosen, can explain more than a fixed fraction of $\|y\|_2^2$ beyond the noise floor; the quantity $g^{-1} + 1 - R_z^2$ appearing in the denominator of Lemma 1.1 is therefore never small. Note that it is here, and only here, that the enlarged model class $\mathcal{X}^+$ of Lemma 1.6 is used.

LEMMA 1.15 (Likelihood comparison). *On $\mathcal{H}_n$, and under (8), every pair $z, z' \in \mathcal{X}$ satisfies*

$$(26) \quad \log \frac{m(z)}{m(z')} \leq \left(\alpha \log d + c_g \right) (|z'| - |z|)^+ + \frac{4\|P_z y\|_2^2}{\sigma_0^2},$$

where $u^+ = \max(u, 0)$.

PROOF. By (4),

$$(27) \quad \log \frac{m(z)}{m(z')} = \underbrace{\frac{1}{2} \log(1+g)(|z'| - |z|)}_{\text{Occam term}} + \underbrace{\frac{n}{2} \log \frac{g^{-1} + 1 - R_{z'}^2}{g^{-1} + 1 - R_z^2}}_{\text{fit term}},$$

where we divided numerator and denominator inside the logarithm by g . The Occam term is at most $(\alpha \log d + c_g)(|z'| - |z|)^+$ by Lemma 1.13, since $\frac{1}{2} \log(1+g) \geq 0$.

For the fit term, if $R_z^2 \leq R_{z'}^2$, the argument of the logarithm is at most 1 and the term is non-positive. Assume therefore $R_z^2 > R_{z'}^2$, and write

$$\frac{g^{-1} + 1 - R_{z'}^2}{g^{-1} + 1 - R_z^2} = 1 + \frac{R_z^2 - R_{z'}^2}{g^{-1} + 1 - R_z^2} \leq 1 + \frac{R_z^2}{1 - R_z^2} = 1 + \frac{\|P_z y\|_2^2}{\|(I - P_z)y\|_2^2},$$

where we dropped $g^{-1} \geq 0$ and $-R_{z'}^2 \leq 0$ and then used $R_z^2 \|y\|_2^2 = \|P_z y\|_2^2$ and $(1 - R_z^2) \|y\|_2^2 = \|(I - P_z)y\|_2^2$. By Lemma 1.14 the denominator is at least $n\sigma_0^2/8$, so, using $\log(1+u) \leq u$,

$$\frac{n}{2} \log \frac{g^{-1} + 1 - R_{z'}^2}{g^{-1} + 1 - R_z^2} \leq \frac{n}{2} \cdot \frac{8\|P_z y\|_2^2}{n\sigma_0^2} = \frac{4\|P_z y\|_2^2}{\sigma_0^2}. \quad \square$$

LEMMA 1.16 (Projection energy). *On $\mathcal{H}_n$, every $z \subseteq x^*$ satisfies*

$$(28) \quad \frac{\|P_z y\|_2^2}{\sigma_0^2} \leq \Psi_\star := 6E + (9\tilde{L} + 3rs^*) \log d.$$

PROOF. Decompose $y = X_S \beta_S^* + X_{S^c} \beta_{S^c}^* + w$ and use $\|a + b + c\|_2^2 \leq 3(\|a\|_2^2 + \|b\|_2^2 + \|c\|_2^2)$ together with the non-expansiveness of P_z :

$$\|P_z y\|_2^2 \leq 3\|X_S \beta_S^*\|_2^2 + 3\|X_{S^c} \beta_{S^c}^*\|_2^2 + 3w^\top P_z w.$$

Since $X_S \beta_S^* = X \beta^* - X_{S^c} \beta_{S^c}^*$ and $\|a - b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$, Assumption 1.2 gives $\|X_S \beta_S^*\|_2^2 \leq 2\sigma_0^2 E + 2\tilde{L}\sigma_0^2 \log d$ and $\|X_{S^c} \beta_{S^c}^*\|_2^2 \leq \tilde{L}\sigma_0^2 \log d$. Finally $|z| \leq s^*$ and the event B_n give $w^\top P_z w \leq r s^* \sigma_0^2 \log d$. Adding up,

$$\frac{\|P_z y\|_2^2}{\sigma_0^2} \leq 6E + 6\tilde{L} \log d + 3\tilde{L} \log d + 3r s^* \log d = \Psi_*. \quad \square$$

The two previous lemmas combine into the single quantity that controls both the length of the tempering schedule and the size of the spectral gap bound of Section 1.5:

$$(29) \quad \Omega_* := (\alpha \log d + c_g) s_+^* + 4\Psi_* = \alpha s_+^* \log d + c_g s_+^* + 24E + (36\tilde{L} + 12r s^*) \log d.$$

COROLLARY 1.1. *On $\mathcal{H}_n$, and under (8), $\log\{m(z)/m(z')\} \leq \Omega_*$ for all $z, z' \subseteq x^*$.*

PROOF. Apply Lemma 1.15 and bound $(|z'| - |z|)^+ \leq s^* \leq s_+^*$ and $\|P_z y\|_2^2 / \sigma_0^2 \leq \Psi_*$ by Lemma 1.16. $\square$

1.3. *Conditions on the design and on the prior.* Every bound in Sections 1.4–1.6 is a comparison between the Occam factor $(1 + g)^{-1/2} = d^{-\alpha + \mathcal{O}(1)}$ and the likelihood gain of Lemma 1.1. It is convenient to name the two objects involved. For $x \in \mathcal{X}$ with $|x| < s_0$ and $j \in [d] \setminus x$, put

$$(30) \quad \psi_j(x) := \log \frac{m(x \cup \{j\})}{m(x)}, \quad G_j(x) := \frac{\|(P_{x \cup \{j\}} - P_x)y\|_2^2}{\sigma_0^2}.$$

We call $\psi_j(x)$ the *log-gain* and $G_j(x)$ the *explained energy* of covariate j in the context x . By (P3),

$$(31) \quad \frac{\pi_\lambda(x \cup \{j\})}{\pi_\lambda(x)} = \exp\{\lambda \psi_j(x) - \kappa \log d\},$$

so that at temperature λ covariate j is worth adding to x exactly when $\lambda \psi_j(x) > \kappa \log d$. We assume from now on that $s^* \geq 1$; if $s^* = 0$, which is the low signal-to-noise regime of Yang, Wainwright and Jordan (2016), then $x^* = \emptyset$, the block decomposition of Section 1.5 is trivial and all the bounds below hold with s^* replaced by 1.

The next lemma sandwiches the log-gain by the explained energy. It is the only place where the data enter the analysis of the spectral gap.

LEMMA 1.17 (Gain transfer). *On $\mathcal{H}_n$, and under (8), for every $x \in \mathcal{X}$ with $|x| < s_0$ and every $j \notin x$,*

$$(32) \quad \frac{n}{4} \min \left\{ \frac{G_j(x)}{b_n}, 1 \right\} - \alpha \log d - c_g \leq \psi_j(x) \leq 4G_j(x) - \alpha \log d, \quad b_n := 3n + \frac{5n}{s^*} + 2E.$$

PROOF. Write $z = x \cup \{j\}$ and use Lemma 1.1:

$$(33) \quad \psi_j(x) = -\frac{1}{2} \log(1 + g) + \frac{n}{2} \log \left(1 + \frac{N}{D} \right), \quad N := \sigma_0^2 G_j(x), \quad D := g^{-1} \|y\|_2^2 + \|(I - P_z)y\|_2^2.$$

Upper bound. Lemma 1.13 gives $-\frac{1}{2}\log(1+g) \leq -\alpha \log d$. Discarding $g^{-1}\|y\|_2^2 \geq 0$ and applying Lemma 1.14 to z yields $D \geq n\sigma_0^2/8$, so, by $\log(1+u) \leq u$,

$$\frac{n}{2} \log\left(1 + \frac{N}{D}\right) \leq \frac{n}{2} \cdot \frac{8\sigma_0^2 G_j(x)}{n\sigma_0^2} = 4G_j(x).$$

Lower bound. Lemma 1.13 gives $-\frac{1}{2}\log(1+g) \geq -\alpha \log d - c_g$. For D we need an upper bound. On D_n , $g^{-1}\|y\|_2^2 \leq 5n\sigma_0^2/s^*$; moreover $\|(I - P_z)y\|_2^2 \leq \|y\|_2^2$ and, by $\|a+b\|_2^2 \leq 2\|a\|_2^2 + 2\|b\|_2^2$ together with the bound $\|w\|_2^2 \leq \frac{3}{2}n\sigma_0^2$ valid on C_n ,

$$\|y\|_2^2 \leq 2\|X\beta^*\|_2^2 + 2\|w\|_2^2 \leq 2\sigma_0^2 E + 3n\sigma_0^2.$$

Hence $D \leq \sigma_0^2 b_n$ and $N/D \geq G_j(x)/b_n$. Finally $\log(1+u) \geq \frac{1}{2}\min(u, 1)$ for all $u \geq 0$: for $u \leq 1$ this is $\log(1+u) \geq u - u^2/2 \geq u/2$, and for $u \geq 1$ it is $\log(1+u) \geq \log 2 \geq 1/2$. Therefore $\frac{n}{2} \log(1 + N/D) \geq \frac{n}{4} \min\{G_j(x)/b_n, 1\}$. $\square$

1.3.1. Null covariates. The first condition says that a covariate outside S never buys enough likelihood to pay for its Occam factor.

ASSUMPTION 1.18 (Null covariates are not worth buying). There is $a_0 \in \mathbb{R}$ such that, for every $x \in \mathcal{X}$ with $|x| < s_0$ and every $j \in S^c \setminus x$,

$$(34) \quad \psi_j(x) \leq (a_0 - \alpha) \log d, \quad \text{equivalently} \quad \frac{m(x \cup \{j\})}{m(x)} \leq d^{a_0 - \alpha}.$$

Assumption 1.18 is a statement about the design alone: it names the largest amount of likelihood that a covariate outside S can buy, measured on the logarithmic scale $\log d$. It says nothing about whether that amount is affordable. Affordability is decided by the prior, and it is temperature-dependent, because by (31) the likelihood gain is damped by λ while the penalty $d^{-\kappa}$ is not: for $x \in \mathcal{X}$ with $|x| < s_0$ and $j \in S^c \setminus x$,

$$(35) \quad \frac{\pi_\lambda(x \cup \{j\})}{\pi_\lambda(x)} = \exp\{\lambda\psi_j(x) - \kappa \log d\} \leq d^{\lambda(a_0 - \alpha) - \kappa} = d^{-\kappa_\lambda}, \quad \kappa_\lambda := \kappa - \lambda(a_0 - \alpha),$$

and we call κ_λ the *effective penalty exponent* at temperature λ . Since there are $\binom{d}{u} \leq d^u$ ways of choosing u null covariates to add, what must be true is that $d^{-\kappa_\lambda}$ beats d , with room to spare. This is the standing condition of the rest of the section.

ASSUMPTION 1.19 (Sparsity budget). There is $c_\kappa > 0$ with $d^{-c_\kappa} \leq 1/2$ such that

$$(36) \quad \kappa_\lambda \geq 1 + c_\kappa \quad \text{for every } \lambda \in [0, 1].$$

REMARK 1.2 (Reading the budget). Since $\lambda \mapsto \kappa_\lambda$ is affine, (36) need only be checked at the two endpoints, where it reads

$$(37) \quad \underbrace{\kappa \geq 1 + c_\kappa}_{\lambda=0} \quad \text{and} \quad \underbrace{\kappa + \alpha \geq 1 + a_0 + c_\kappa}_{\lambda=1}.$$

At $\lambda = 0$ the target is the prior, the likelihood is switched off, and only the sparsity exponent κ is available to beat the d ways of adding a covariate. At $\lambda = 1$ the full gain d^{a_0} is in play, and both the sparsity prior and the Occam factor of the g -prior contribute, through κ and α respectively. The second inequality has the same shape as Assumption 1.4, which reads $\kappa + \alpha \geq c_\alpha(L + \tilde{L}) + 2$; the two are compared in Remark 1.3.

Suppose the first inequality in (37) holds and the second fails, so that $\kappa_\lambda \geq 1 + c_\kappa$ holds exactly on $[0, \lambda_c]$ with

$$(38) \quad \lambda_c := \min \left\{ 1, \frac{\kappa - 1 - c_\kappa}{(a_0 - \alpha)_+} \right\} < 1.$$

Then every statement below that is asserted for $\lambda \in [0, 1]$ holds instead for $\lambda \in [0, \lambda_c]$, with the same proof: the only uses of $\lambda \leq 1$ are through (36) and through the monotonicity steps in Lemmas 1.23 and 1.30, all of which are valid a fortiori on the shorter interval. So λ_c is the temperature beyond which the analysis of this section is silent, and it is computable from (κ, α, a_0) .

The next two propositions bound a_0 .

PROPOSITION 1.1 (Overfitted contexts). *On $\mathcal{H}_n$, and under (8), inequality (34) holds with $a_0 = 8(L + \tilde{L})$ for every $x \supseteq x^*$ with $|x| < s_0$ and every $j \in S^c \setminus x$.*

PROOF. Write $z = x \cup \{j\}$ and $\Pi := P_z - P_x$, an orthogonal projection of rank one. Since $S \subseteq x$, the vector $X_S \beta_S^*$ lies in the range of P_x , so $\Pi X_S \beta_S^* = \Pi P_x X_S \beta_S^* = 0$ and

$$\sigma_0^2 G_j(x) = \|\Pi y\|_2^2 = \|\Pi(X_{S^c} \beta_{S^c}^* + w)\|_2^2 \leq 2\|X_{S^c} \beta_{S^c}^*\|_2^2 + 2w^\top \Pi w.$$

By Assumption 1.2 the first term is at most $2\tilde{L}\sigma_0^2 \log d$; by the event A_n , applied to the nested pair $x \subset z$ of $\mathcal{X}^+$ with $|z| - |x| = 1$, the second is at most $2L\sigma_0^2 \log d$. Hence $G_j(x) \leq 2(L + \tilde{L}) \log d$ and the upper bound of Lemma 1.17 gives $\psi_j(x) \leq 8(L + \tilde{L}) \log d - \alpha \log d$. $\square$

Proposition 1.1 recovers the exponent $8(L + \tilde{L}) - \alpha$ of the overfitted computation in the proof of Theorem 2 of Yang, Wainwright and Jordan (2016). What it does not cover is an *underfitted* context, in which a null covariate may act as a proxy for a missing influential one. The obstruction is exactly the quantity

$$(39) \quad \Xi := \max_{x \in \mathcal{X}, |x| < s_0} \max_{j \in S^c \setminus x} \frac{\|(P_{x \cup \{j\}} - P_x) X_S \beta_S^*\|_2^2}{\sigma_0^2},$$

which we call the *proxy energy* of the design: it is the largest amount of true signal that a single null column can explain, whatever the covariates already in the model.

PROPOSITION 1.2 (General contexts). *On $\mathcal{H}_n$, and under (8), inequality (34) holds with*

$$a_0 = 12(L + \tilde{L}) + \frac{12\Xi}{\log d}.$$

Moreover $\Xi \leq 2E + 2\tilde{L} \log d$ always, and $\Xi = 0$ whenever the null columns are orthogonal to the influential ones, that is whenever $X_j^\top X_k = 0$ for all $j \in S^c$ and $k \in S$.

PROOF. With $\Pi = P_{x \cup \{j\}} - P_x$ as above, decompose $y = X_S \beta_S^* + X_{S^c} \beta_{S^c}^* + w$ and use $\|a + b + c\|_2^2 \leq 3(\|a\|_2^2 + \|b\|_2^2 + \|c\|_2^2)$:

$$\sigma_0^2 G_j(x) \leq 3\|\Pi X_S \beta_S^*\|_2^2 + 3\|X_{S^c} \beta_{S^c}^*\|_2^2 + 3w^\top \Pi w \leq 3\sigma_0^2 \Xi + 3\tilde{L}\sigma_0^2 \log d + 3L\sigma_0^2 \log d,$$

using (39), Assumption 1.2 and the event A_n . The upper bound of Lemma 1.17 then gives $\psi_j(x) \leq 12\Xi + 12(L + \tilde{L}) \log d - \alpha \log d$.

For the two claims about Ξ , write $u := (I - P_x)X_j$, so that Π is the orthogonal projection onto $\text{span}(u)$ and

$$(40) \quad \|\Pi X_S \beta_S^*\|_2^2 = \frac{\langle u, X_S \beta_S^* \rangle^2}{\|u\|_2^2}.$$

Since $\Pi = \Pi(I - P_x)$, we have $\langle u, X_S \beta_S^* \rangle = \langle u, (I - P_x)X_S \beta_S^* \rangle$, so Cauchy–Schwarz gives $\|\Pi X_S \beta_S^*\|_2^2 \leq \|(I - P_x)X_S \beta_S^*\|_2^2 \leq \|X_S \beta_S^*\|_2^2 \leq 2\sigma_0^2 E + 2\tilde{L}\sigma_0^2 \log d$, the last step as in the proof of Lemma 1.16.

Finally suppose $X_j^\top X_k = 0$ for $j \in S^c$, $k \in S$, and split $x = x_1 \sqcup x_0$ with $x_1 \subseteq S$ and $x_0 \subseteq S^c$. The two column blocks are orthogonal, so $P_x = P_{x_1} + P_{x_0}$. For $j \in S^c \setminus x$ we get $P_{x_1} X_j = 0$, hence $u = X_j - P_{x_0} X_j$ lies in $\text{span}(X_{S^c})$, which is orthogonal to $\text{span}(X_S) \ni X_S \beta_S^*$. Thus $\langle u, X_S \beta_S^* \rangle = 0$ and $\Xi = 0$ by (40). $\square$

REMARK 1.3. *The two bounds on Ξ delimit what the sparsity budget (37) costs, and Ξ is the only quantity that separates the two regimes.*

If no null column can proxy for the signal — orthogonality is the extreme case, and Ξ is in general controlled by the quantity $\omega(X)$ of Assumption 1.5 — then $\Xi = \mathcal{O}(\log d)$, hence $a_0 = \mathcal{O}(L + \tilde{L})$ and the budget reads $\kappa + \alpha \gtrsim L + \tilde{L}$. This is exactly the shape of Assumption 1.4, and is implied by it as soon as the universal constant c_α there is at least 13. In this regime the analysis of the present section asks for nothing that Yang, Wainwright and Jordan (2016) do not already assume.

At the other extreme, the trivial bound $\Xi \leq 2E + 2\tilde{L} \log d$ together with $E = \mathcal{O}(s^ \log d)$ (Remark 1.5) gives only $a_0 = \mathcal{O}(s^*)$, so the budget demands $\kappa + \alpha = \Omega(s^*)$, that is $g = d^{\Omega(s^*)}$ or a sparsity prior of comparable strength. Assumption 1.18 together with Assumption 1.19 is therefore a genuine requirement, not a consequence of Assumptions 1.2–1.5: it is the price of controlling the sampler at every temperature rather than at $\lambda = 1$ only, where the posterior itself suppresses underfitted states.*

1.3.2. *Influential covariates.* The second condition is the mirror image of the first: an influential covariate is always worth buying, whatever the other influential covariates already present. For $j \in S$ set

$$(41) \quad \psi_j^- := \min_{x \subseteq S \setminus \{j\}} \psi_j(x), \quad \psi_j^+ := \max_{x \subseteq S \setminus \{j\}} \psi_j(x), \quad \Delta_{\text{int}} := \max_{j \in S} (\psi_j^+ - \psi_j^-).$$

We call Δ_{int} the *interaction spread*: it measures by how much the gain from adding an influential covariate depends on which other influential covariates are already in the model. Note that only subsets of S occur in (41); the behaviour of ψ_j in contexts containing null covariates is never needed.

ASSUMPTION 1.20 (Influential covariates are worth buying). *There is $a_1 > 0$ with $d^{a_1} \geq 2s^*$ such that $\psi_j^- \geq (\kappa + a_1) \log d$ for every $j \in S$.*

PROPOSITION 1.3. *On $\mathcal{H}_n$, and under (8), Assumption 1.20 holds as soon as*

$$(42) \quad \min_{j \in S} \min_{x \subseteq S \setminus \{j\}} G_j(x) \geq \frac{4b_n}{n} \left\{ (\alpha + \kappa + a_1) \log d + c_g \right\} \quad \text{and} \quad \frac{4}{n} \left\{ (\alpha + \kappa + a_1) \log d + c_g \right\} \leq 1.$$

In particular, if the columns of X are orthogonal and the threshold satisfies the $\beta_{\min}$ condition $C_\beta^2 \geq c_0(L + \tilde{L} + \alpha + \kappa)\sigma_0^2 \log d/n$ with $c_0 \geq 4r$, then $\min_{j,x} G_j(x) \geq nC_\beta^2/(4\sigma_0^2) \geq \frac{c_0}{4}(L + \tilde{L} + \alpha + \kappa) \log d$.

PROOF. The second condition in (42) makes the minimum in the lower bound of Lemma 1.17 equal to $G_j(x)/b_n$ whenever the first condition holds, so that

$$\psi_j(x) \geq \frac{n}{4} \cdot \frac{G_j(x)}{b_n} - \alpha \log d - c_g \geq (\alpha + \kappa + a_1) \log d + c_g - \alpha \log d - c_g = (\kappa + a_1) \log d$$

for every $j \in S$ and $x \subseteq S \setminus \{j\}$, which is Assumption 1.20.

For the orthogonal case, $P_{x \cup \{j\}} - P_x$ is the projection onto $\text{span}(X_j)$ for every $x \not\ni j$, so $\sigma_0^2 G_j(x) = \langle X_j, y \rangle^2 / n$, independently of x . Now $\langle X_j, y \rangle = n\beta_j^* + \langle X_j, w \rangle$ because the columns are orthogonal and $\|X_j\|_2^2 = n$, while $\langle X_j, w \rangle^2 / n = w^\top P_{\{j\}} w \leq r\sigma_0^2 \log d$ on the event B_n . Hence

$$\sigma_0 \sqrt{G_j(x)} \geq \sqrt{n} |\beta_j^*| - \sigma_0 \sqrt{r \log d} \geq \sqrt{n} C_\beta - \sigma_0 \sqrt{r \log d} \geq \frac{1}{2} \sqrt{n} C_\beta,$$

the last step because $\sqrt{n} C_\beta \geq \sigma_0 \sqrt{c_0(L + \tilde{L} + \alpha + \kappa) \log d} \geq 2\sigma_0 \sqrt{r \log d}$ when $c_0 \geq 4r$. $\square$

The two assumptions together locate the maximum of m , which is what Section 1.4 needs. Note that Assumption 1.18 alone does not make x^* the maximiser, since it allows m to increase, by a factor at most $d^{a_0 - \alpha}$, when a null covariate is added; what it does give is that m cannot exceed $m(x^*)$ by more than the accumulation of those factors.

LEMMA 1.21 (Near-maximality of x^*). *Under Assumptions 1.18 and 1.20, every $x \in \mathcal{X}$ satisfies*

$$(43) \quad m(x) \leq d^{(a_0 - \alpha)_+ |x \cap S^c|} m(x^*) \leq d^{(a_0 - \alpha)_+ s_0} m(x^*),$$

with $u^+ = \max(u, 0)$ as in Lemma 1.15. In particular $m(x^*) = \max_{x \in \mathcal{X}} m(x)$ when $a_0 \leq \alpha$.

PROOF. Fix $x \in \mathcal{X}$, set $u := |x \cap S^c|$ and enumerate $x \cap S^c = \{j_1, \dots, j_u\}$. Deleting these indices one at a time produces

$$x = x_0 \supset x_1 \supset \dots \supset x_u = x \cap S, \quad x_i = x_{i-1} \setminus \{j_i\},$$

all of which lie in $\mathcal{X}$, since deletion only decreases the cardinality, and all of which satisfy $|x_i| = |x| - i \leq s_0 - 1 < s_0$ for $i \geq 1$. Since $x_{i-1} = x_i \cup \{j_i\}$ with $j_i \in S^c \setminus x_i$, Assumption 1.18 applies at the state x_i and gives $m(x_{i-1}) \leq d^{a_0 - \alpha} m(x_i)$. Multiplying these u inequalities and using $d^{a_0 - \alpha} \leq d^{(a_0 - \alpha)_+}$,

$$(44) \quad m(x) \leq d^{u(a_0 - \alpha)} m(x \cap S) \leq d^{u(a_0 - \alpha)_+} m(x \cap S).$$

Next enumerate $S \setminus x = \{k_1, \dots, k_v\}$ and add these indices one at a time, producing $x \cap S = z_0 \subset z_1 \subset \dots \subset z_v = x^*$, all of them subsets of S and hence of cardinality at most $s^* < s_0$. At each step $z_{i-1} \subseteq S \setminus \{k_i\}$ with $k_i \in S$, so (41) and Assumption 1.20 give

$$\psi_{k_i}(z_{i-1}) \geq \psi_{k_i}^- \geq (\kappa + a_1) \log d > 0, \quad \text{that is} \quad m(z_{i-1}) \leq m(z_i).$$

Hence $m(x \cap S) \leq m(x^*)$, which with (44) and $u \leq |x| \leq s_0$ gives (43). If $a_0 \leq \alpha$ then $(a_0 - \alpha)_+ = 0$ and (43) reads $m(x) \leq m(x^*)$. $\square$

1.4. *How many distributions are needed.* Theorem ?? produces a schedule satisfying Assumption ?? when $\mathcal{X} = \mathbb{R}^d$ and the intermediate densities are log-concave and smooth. Neither hypothesis is available here. What replaces them is that $\lambda \mapsto \pi_\lambda$ is an exponential family in λ with sufficient statistic $\log m$, so that the χ^2 -divergence between two of its members has a closed form in terms of the cumulant generating function

$$(45) \quad \varphi(\lambda) := \log Z(\lambda) = \log \sum_{x \in \mathcal{X}} p(x) m(x)^\lambda,$$

which is smooth and convex on $\mathbb{R}$. Everything is then controlled by the variance function

$$(46) \quad V(\lambda) := \text{Var}_{\pi_\lambda}[\log m] = \varphi''(\lambda),$$

and, after integration in λ , by the symmetrised Kullback–Leibler divergence between the endpoints of the path.

1.4.1. A schedule from the variance function.

PROPOSITION 1.4. *Let $\mathcal{X}$ be a finite set, $m : \mathcal{X} \rightarrow (0, \infty)$, and ν a probability measure with $\nu(x) > 0$ for every $x \in \mathcal{X}$. Let $\pi_\lambda(x) = m(x)^\lambda \nu(x) / Z(\lambda)$ with $Z(\lambda) = \nu(m^\lambda)$, let V be as in (46) and let $\bar{V} := \max_{\lambda \in [0, 2]} V(\lambda)$.*

(i) *For all $\lambda \geq 0$ and $h \geq 0$,*

$$(47) \quad 1 + \chi^2(\pi_{\lambda+h} \mid \pi_\lambda) = \frac{Z(\lambda+2h)Z(\lambda)}{Z(\lambda+h)^2}, \quad \log\{1 + \chi^2(\pi_{\lambda+h} \mid \pi_\lambda)\} = \int_0^h \int_0^h V(\lambda+u+v) \, du \, dv.$$

(ii) *Set $\lambda_{-1} = 0$ and, for $t \geq 0$, $\lambda_t = \lambda_{t-1} + \Delta_t$, where Δ_t is the largest number in $(0, 1 - \lambda_{t-1}]$ such that*

$$(48) \quad \Delta_t^2 \max\{V(\lambda) : \lambda \in [\lambda_{t-1}, \lambda_{t-1} + 2\Delta_t]\} \leq \log 2.$$

Then Δ_t exists, the recursion reaches $\lambda_T = 1$ in finitely many steps, the resulting sequence $(\pi_t)_{t=-1}^T$ satisfies Assumption ??, and

$$(49) \quad T + 1 \leq \left\lceil \sqrt{\bar{V} / \log 2} \right\rceil.$$

(iii) *Suppose in addition that there is $C_V \geq 1$ such that*

$$(50) \quad V(\mu) \leq C_V V(\lambda) \quad \text{whenever } \lambda \in [0, 1], \mu \in [0, 2], |\mu - \lambda| \leq 2\sqrt{\log 2 / V(\lambda)}.$$

Then the schedule of ii satisfies

$$(51) \quad T \leq \frac{C_V}{\sqrt{\log 2}} \int_0^1 \sqrt{V(\lambda)} \, d\lambda \leq \frac{C_V}{\sqrt{\log 2}} \sqrt{\mathcal{J}}, \quad \mathcal{J} := \text{KL}(\nu \mid \pi_1) + \text{KL}(\pi_1 \mid \nu).$$

PROOF. Throughout, $\mathcal{X}$ is finite and $m > 0$, so $Z(\lambda) \in (0, \infty)$ for every real λ , all the π_λ have full support and are mutually absolutely continuous, and $\varphi = \log Z$ is real analytic with

$$(52) \quad \varphi'(\lambda) = \mathbb{E}_{\pi_\lambda}[\log m], \quad \varphi''(\lambda) = \text{Var}_{\pi_\lambda}[\log m] = V(\lambda) \geq 0.$$

These are the standard exponential-family identities: differentiating $Z(\lambda) = \sum_x \nu(x) e^{\lambda \log m(x)}$ under the finite sum gives $Z'(\lambda) = \sum_x \nu(x) m(x)^\lambda \log m(x) = Z(\lambda) \mathbb{E}_{\pi_\lambda}[\log m]$, whence the first identity, and differentiating once more gives the second.

Proof of i. By definition of the χ^2 -divergence between two measures with the same support,

$$1 + \chi^2(\pi_{\lambda+h} \mid \pi_\lambda) = \sum_{x \in \mathcal{X}} \frac{\pi_{\lambda+h}(x)^2}{\pi_\lambda(x)} = \sum_x \frac{\nu(x)^2 m(x)^{2\lambda+2h}}{Z(\lambda+h)^2} \cdot \frac{Z(\lambda)}{\nu(x)m(x)^\lambda} = \frac{Z(\lambda)}{Z(\lambda+h)^2} \sum_x \nu(x)m(x)^{\lambda+2h},$$

and the remaining sum is $Z(\lambda+2h)$. This is the first identity. Taking logarithms, the left-hand side of the second identity equals

$$\Phi(h) := \varphi(\lambda+2h) + \varphi(\lambda) - 2\varphi(\lambda+h),$$

and we must show that $\Phi = \Psi$, where $\Psi(h) := \int_0^h \int_0^h V(\lambda+u+v) du dv$. Both vanish at $h=0$. Differentiating Ψ by the Leibniz rule, once for the upper limit in u and once for the upper limit in v ,

$$\Psi'(h) = \int_0^h V(\lambda+h+v) dv + \int_0^h V(\lambda+u+h) du = 2 \int_0^h V(\lambda+h+u) du = 2\{\varphi'(\lambda+2h) - \varphi'(\lambda+h)\},$$

by (52), while $\Phi'(h) = 2\varphi'(\lambda+2h) - 2\varphi'(\lambda+h)$. The two derivatives agree, so $\Phi \equiv \Psi$.

Proof of ii. Fix t and write $a = \lambda_{t-1} \in [0, 1]$ and

$$M(\Delta) := \max\{V(\lambda) : \lambda \in [a, a+2\Delta]\}, \quad h(\Delta) := \Delta^2 M(\Delta), \quad \Delta \in [0, 1-a].$$

These are well defined: $a+2\Delta \leq a+2(1-a) = 2-a \leq 2$, so the interval $[a, a+2\Delta]$ lies in $[0, 2]$, on which V is continuous by (52) and hence attains its maximum.

M is non-decreasing, because the intervals are nested, and it is continuous. For continuity, let $0 \leq \Delta < \Delta' \leq 1-a$ and let $\lambda' \in [a, a+2\Delta']$ attain $M(\Delta')$; the point $\lambda := \min\{\lambda', a+2\Delta\}$ lies in $[a, a+2\Delta]$ and satisfies $|\lambda - \lambda'| \leq 2(\Delta' - \Delta)$, so

$$0 \leq M(\Delta') - M(\Delta) \leq V(\lambda') - V(\lambda) \leq \omega_V(2(\Delta' - \Delta)),$$

where ω_V is the modulus of continuity of V on $[0, 2]$, which tends to 0 at 0 because V is uniformly continuous there. Hence h is continuous and non-decreasing on $[0, 1-a]$, with $h(0) = 0 < \log 2$.

Consequently $\mathcal{A} := \{\Delta \in [0, 1-a] : h(\Delta) \leq \log 2\}$ is a non-empty closed subinterval of $[0, 1-a]$ containing 0, and $\Delta_t := \max \mathcal{A}$ exists. If $\bar{V} = 0$ then $V \equiv 0$ on $[0, 2]$, so $\log m$ is π_λ -a.s. constant, hence constant on $\mathcal{X}$; then $\pi_\lambda = \nu$ for all λ , $h \equiv 0$, $\Delta_0 = 1$ and $T = 0$. Assume therefore $\bar{V} > 0$. Since $h(\Delta) \leq \Delta^2 \bar{V}$, every $\Delta \leq \sqrt{\log 2 / \bar{V}}$ lies in $\mathcal{A}$, so

$$(53) \quad \Delta_t \geq \min\left\{1 - \lambda_{t-1}, \sqrt{\log 2 / \bar{V}}\right\}.$$

Consequently the schedule increases by at least $\sqrt{\log 2 / \bar{V}}$ at each step until it reaches 1, so it does reach 1 after at most $\lceil \sqrt{\bar{V} / \log 2} \rceil$ steps. The steps of the schedule are $\Delta_0, \dots, \Delta_T$, that is $T+1$ of them, which gives (49).

It remains to check Assumption ??, that is $1 + \chi^2(\pi_t \mid \pi_{t-1}) \leq 2$ for every t . Apply the second identity of (47) with $\lambda = \lambda_{t-1}$ and $h = \Delta_t$. In the double integral, $u+v$ ranges over $[0, 2\Delta_t]$, so the integrand is evaluated only on $[\lambda_{t-1}, \lambda_{t-1} + 2\Delta_t]$ and

$$\log\{1 + \chi^2(\pi_t \mid \pi_{t-1})\} \leq \Delta_t^2 \max\{V(\lambda) : \lambda \in [\lambda_{t-1}, \lambda_{t-1} + 2\Delta_t]\} \leq \log 2$$

by (48).

Proof of iii. We first improve (53) into a local bound. Fix t and put $a = \lambda_{t-1}$. If $\Delta \leq \sqrt{\log 2 / (C_V V(a))}$ then, since $C_V \geq 1$, $2\Delta \leq 2\sqrt{\log 2 / V(a)}$, so every $\mu \in [a, a+2\Delta]$

is within $2\sqrt{\log 2/V(a)}$ of $a \in [0, 1]$ and (50) gives $V(\mu) \leq C_V V(a)$. Hence $h(\Delta) \leq \Delta^2 C_V V(a) \leq \log 2$, so Δ is admissible and

$$(54) \quad \Delta_t \geq \sqrt{\frac{\log 2}{C_V V(\lambda_{t-1})}} \quad \text{whenever } \sqrt{\log 2/(C_V V(\lambda_{t-1}))} \leq 1 - \lambda_{t-1},$$

Conversely, if the displayed condition fails then $\Delta = 1 - \lambda_{t-1}$ is itself admissible, so $\Delta_t = 1 - \lambda_{t-1}$, $\lambda_t = 1$ and $t = T$. Hence (54) holds for every $t < T$: indeed $t < T$ means $\lambda_t < 1$, that is $\Delta_t < 1 - \lambda_{t-1}$, which excludes the second alternative. Counting the steps other than the last,

$$(55) \quad T = \sum_{t=0}^{T-1} 1 \leq \sum_{t=0}^{T-1} \Delta_t \sqrt{\frac{C_V V(\lambda_{t-1})}{\log 2}},$$

the last step of the schedule being counted separately as the single unit already subtracted on the left. Now fix t and $\lambda \in [\lambda_{t-1}, \lambda_t]$. Since $\lambda \in [\lambda_{t-1}, \lambda_{t-1} + 2\Delta_t]$, the constraint (48) gives $\Delta_t^2 V(\lambda) \leq \log 2$, that is $|\lambda - \lambda_{t-1}| \leq \Delta_t \leq \sqrt{\log 2/V(\lambda)}$; applying (50) with this λ and $\mu = \lambda_{t-1}$ yields $V(\lambda_{t-1}) \leq C_V V(\lambda)$, hence $\sqrt{C_V V(\lambda_{t-1})} \leq C_V \sqrt{V(\lambda)}$. Integrating this over $\lambda \in [\lambda_{t-1}, \lambda_t]$, an interval of length Δ_t ,

$$\Delta_t \sqrt{C_V V(\lambda_{t-1})} \leq C_V \int_{\lambda_{t-1}}^{\lambda_t} \sqrt{V(\lambda)} d\lambda,$$

and summing over $t \leq T - 1$ in (55), the intervals $[\lambda_{t-1}, \lambda_t]$ being disjoint and contained in $[0, 1]$, gives the first inequality of (51).

For the second, Cauchy–Schwarz on $[0, 1]$ gives $\int_0^1 \sqrt{V} \leq (\int_0^1 V)^{1/2}$, and by (52),

$$\int_0^1 V(\lambda) d\lambda = \varphi'(1) - \varphi'(0) = \mathbb{E}_{\pi_1}[\log m] - \mathbb{E}_{\nu}[\log m].$$

Since $d\pi_1/d\nu = m/Z(1)$, we have $\text{KL}(\pi_1 | \nu) = \mathbb{E}_{\pi_1}[\log m] - \log Z(1)$ and $\text{KL}(\nu | \pi_1) = \log Z(1) - \mathbb{E}_{\nu}[\log m]$, whose sum is $\mathcal{J}$. $\square$

REMARK 1.4. *Three comments on the schedule (48). First, it is an explicit schedule, not an adaptive one: V is a deterministic function of the data and could in principle be evaluated, although in practice one would use the adaptive effective-sample-size rule of Schafer and Chopin (2013), for which (48) provides the theoretical justification. Second, the constraint looks ahead to $\lambda_{t-1} + 2\Delta_t$ rather than to $\lambda_t = \lambda_{t-1} + \Delta_t$: this is forced by the first identity in (47), in which the χ^2 -divergence at $(\lambda, \lambda + h)$ involves Z at $\lambda + 2h$. It is the discrete analogue of the fact that χ^2 control requires a bound on the density ratio squared. Third, the regularity condition (50) is what turns the crude bound (49), which pays the worst variance along the whole path, into the integral bound (51). It holds with $C_V = \mathcal{O}(1)$ whenever $\log V$ is Lipschitz on $[0, 2]$ with constant $\mathcal{O}(\sqrt{V})$, which is the generic situation.*

1.4.2. Bounding the symmetrised divergence. It remains to bound $\mathcal{J}$ for the path (10). This is where the data enter, through Section 1.2.

COROLLARY 1.2. *Assume 1.18, 1.20 and 1.19, place yourself on $\mathcal{H}_n$, and assume (8). Then, for the path (10),*

$$(56) \quad \mathcal{J} \leq \mathcal{J}_* := \Omega_* + (a_0 - \alpha)_+ s_0 \log d,$$

with Ω_* as in (29). In particular $\mathcal{J} \leq \Omega_*$ when $a_0 \leq \alpha$.

PROOF. By the computation at the end of the proof of Proposition 1.4, $\mathcal{J} = \mathbb{E}_\pi[\log m] - \mathbb{E}_\nu[\log m]$ with $\pi = \pi_1$. Bounding the first expectation by the maximum of its integrand and applying Lemma 1.21,

$$(57) \quad \mathbb{E}_\pi[\log m] \leq \max_{x \in \mathcal{X}} \log m(x) \leq \log m(x^*) + (a_0 - \alpha)_+ s_0 \log d,$$

so that

$$(58) \quad \mathcal{J} \leq (a_0 - \alpha)_+ s_0 \log d + \mathbb{E}_{x \sim \nu} \left[\log \frac{m(x^*)}{m(x)} \right].$$

Apply Lemma 1.15 with $z = x^*$ and $z' = x$, then Lemma 1.16 with $z = x^*$:

$$(59) \quad \log \frac{m(x^*)}{m(x)} \leq (\alpha \log d + c_g)(|x| - s^*)^+ + 4\Psi_* \leq (\alpha \log d + c_g)|x| + 4\Psi_*.$$

It remains to bound $\mathbb{E}_\nu[|x|]$. Since $\nu(x) \propto d^{-\kappa|x|}$ and there are $\binom{d}{k} \leq d^k$ states with $|x| = k$,

$$\frac{\nu(|x| = k)}{\nu(|x| = 0)} \leq \binom{d}{k} d^{-\kappa k} \leq d^{k(1-\kappa)} \leq d^{-c_\kappa k} \leq 2^{-k},$$

using $\kappa \geq 1 + c_\kappa$, which is the $\lambda = 0$ half of (37), and $d^{-c_\kappa} \leq 1/2$. Hence, with $\nu(|x| = 0) \leq 1$,

$$(60) \quad \mathbb{E}_\nu[|x|] = \sum_{k \geq 1} k \nu(|x| = k) \leq \sum_{k \geq 1} k 2^{-k} = 2 \leq s_+^*.$$

Substituting into (59) and (58) gives

$$\mathcal{J} \leq (a_0 - \alpha)_+ s_0 \log d + (\alpha \log d + c_g) s_+^* + 4\Psi_* = (a_0 - \alpha)_+ s_0 \log d + \Omega_* = \mathcal{J}_*$$

by (29). $\square$

REMARK 1.5. *The horizon is therefore*

$$(61) \quad T \leq \frac{C_V}{\sqrt{\log 2}} \sqrt{\mathcal{J}_*}, \quad \mathcal{J}_* = \Omega_* + (a_0 - \alpha)_+ s_0 \log d.$$

To read off its size, suppose the signal sits at the detection threshold, $|\beta_j^*| \asymp C_\beta$ for $j \in S$ with $C_\beta^2 \asymp (L + \tilde{L} + \alpha + \kappa) \sigma_0^2 \log d / n$, and suppose the design satisfies an upper restricted eigenvalue condition, $\max_{|x| \leq s_0} \lambda_{\max}(n^{-1} X_x^\top X_x) \leq C_M$, the companion of Assumption 1.3. Then $\|X_S \beta_S^*\|_2^2 \leq C_M n \|\beta_S^*\|_2^2 \asymp C_M n s^* C_\beta^2$, so

$$E = \mathcal{O}((L + \tilde{L} + \alpha + \kappa) s^* \log d), \quad \mathcal{J}_* = \mathcal{O}((\alpha + L + \tilde{L} + \kappa) s_+^* \log d + (a_0 - \alpha)_+ s_0 \log d), \quad T = \mathcal{O}(\sqrt{s_+^* \log d})$$

with $\kappa, \alpha, L, \tilde{L}, r, C_V = \mathcal{O}(1)$ and $a_0 \leq \alpha$; if instead $a_0 > \alpha$, the second term of $\mathcal{J}_*$ adds $\mathcal{O}(\sqrt{(a_0 - \alpha) s_0 \log d})$ to the horizon, which by Assumption 1.19 is at most $\mathcal{O}(\sqrt{(\kappa - 1) s_0 \log d})$. Two features are worth noting. The horizon is governed by the true sparsity s^* , not by the truncation level s_0 : the prior ν already concentrates on models of size $\mathcal{O}(1)$ by (60), so the path has only to travel a distance s^* in model space. And n does not appear except through E ; at the detection threshold the growth of n is exactly compensated by the shrinking of C_β . Without the upper eigenvalue condition, only the a priori bound $E \leq g \log d$ of Assumption 1.2 is available, and $T = \mathcal{O}(d^\alpha \sqrt{\log d})$; the upper eigenvalue condition is thus doing real work here, and we state it explicitly rather than leaving it implicit in the phrase “at the detection threshold”.

1.5. *The spectral gap at every temperature.* Theorem 2 of [Yang, Wainwright and Jordan \(2016\)](#) bounds the spectral gap of K_1 , the kernel at the posterior. Assumption ?? needs a bound for all $T + 1$ kernels of the path (10), and Theorem ?? needs one for τ_β as well. This subsection gives such a bound; Section 1.6 sharpens it.

1.5.1. *Why the ensemble of Yang, Wainwright and Jordan does not transfer.* Their proof uses a canonical path ensemble on the whole of $\mathcal{X}$ directed at x^* , in the sense of Definition 1.11 with $z_\bullet = x^*$: from an overfitted state one deletes a null covariate, from an underfitted state one adds an influential one. The congestion bound then reads $\varrho \leq 4d \max_z \pi_\lambda[\text{Prec}(z)]/\pi_\lambda(z)$, and it is finite because each step of the map increases π_1 by a factor d^2 or more, so that $\text{Prec}(z)$ has geometrically decreasing mass.

At $\lambda = 0$ this fails completely. There $\pi_0 = p$ is the sparsity prior, which puts mass $\propto d^{-\kappa|x|}$ on x and is therefore maximised at $x = \emptyset$; a map directed at x^* moves *downhill* at every step of the second kind, each step costing a factor $d^{-\kappa}$. The set $\text{Prec}(x^*)$ is then all of $\mathcal{X}$, whose π_0 -mass exceeds $\pi_0(x^*)$ by a factor $d^{\kappa s^*}$ at least, and the resulting bound is worse than trivial. The same happens at any λ small enough that $\lambda\psi_j^- < \kappa \log d$, that is on an initial segment of the path whose length does not go to zero.

There are two ways out, and we take both. The first, in this subsection, avoids path ensembles on $\mathcal{X}$ altogether: it splits $\mathcal{X}$ into blocks in which the influential covariates are frozen, handles each block by a path ensemble that only ever deletes null covariates — a direction in which π_λ increases at every temperature, by (P3) — and treats the 2^{s^*} blocks by brute force. The second, in Section 1.6, keeps a path ensemble on $\mathcal{X}$ but directs it at the mode of π_λ rather than at x^* .

1.5.2. *The block decomposition and mass ratios.* Recall $S^c = [d] \setminus S$. For $R \subseteq S$ set

$$(62) \quad B_R := \{x \in \mathcal{X} : x \cap S = R\} = \{R \cup U : U \subseteq S^c, |U| \leq s_0 - |R|\},$$

so that $\mathcal{X} = \bigsqcup_{R \subseteq S} B_R$ is a partition into 2^{s^*} non-empty blocks. Inside a block the influential covariates are frozen and only null ones move; between blocks only influential ones move. We write $\bar{\pi}_\lambda(R) := \pi_\lambda(B_R)$ for the block masses.

LEMMA 1.22 (Mass ratios). *Assume 1.18 and 1.19. Then for every $\lambda \in [0, 1]$:*

(i) *for every $x \in \mathcal{X}$ and every $W \subseteq S^c \setminus x$ with $|x| + |W| \leq s_0$,*

$$(63) \quad \frac{\pi_\lambda(x \cup W)}{\pi_\lambda(x)} \leq d^{-\kappa_\lambda |W|} \leq d^{-(1+c_\kappa)|W|},$$

(ii) *for every $x \in \mathcal{X}$,*

$$(64) \quad \sum_{\substack{W \subseteq S^c \setminus x \\ |x| + |W| \leq s_0}} \frac{\pi_\lambda(x \cup W)}{\pi_\lambda(x)} \leq 2;$$

(iii) *for every $R \subseteq S$, $\pi_\lambda(R) \leq \bar{\pi}_\lambda(R) \leq 2\pi_\lambda(R)$.*

PROOF. i Write $W = \{j_1, \dots, j_u\}$ and $W_i = \{j_1, \dots, j_i\}$, so that $W_0 = \emptyset$ and $W_u = W$. Each $x \cup W_i$ lies in $\mathcal{X}$, and for $i < u$ we have $|x \cup W_i| = |x| + i \leq s_0 - (u - i) < s_0$, so Assumption 1.18 applies at each of the u steps, in the form (35). Telescoping,

$$\frac{\pi_\lambda(x \cup W)}{\pi_\lambda(x)} = \prod_{i=1}^u \frac{\pi_\lambda(x \cup W_i)}{\pi_\lambda(x \cup W_{i-1})} = d^{-\kappa u} \prod_{i=1}^u \exp\{\lambda\psi_{j_i}(x \cup W_{i-1})\} \leq d^{-\kappa u} d^{\lambda u(a_0 - \alpha)} = d^{-\kappa_\lambda u},$$

by the definition of κ_λ in (35); and $d^{-\kappa_\lambda u} \leq d^{-(1+c_\kappa)u}$ by Assumption 1.19.

ii Group the subsets W by their cardinality w ; there are at most $\binom{d}{w} \leq d^w$ of them, so by (63),

$$\sum_W \frac{\pi_\lambda(x \cup W)}{\pi_\lambda(x)} \leq \sum_{w \geq 0} d^w d^{-\kappa_\lambda w} = \sum_{w \geq 0} (d^{1-\kappa_\lambda})^w \leq \sum_{w \geq 0} (d^{-c_\kappa})^w \leq \frac{1}{1-1/2} = 2,$$

the third step by Assumption 1.19 and the fourth by $d^{-c_\kappa} \leq 1/2$.

iii Take $x = R$ in ii: the subsets $W \subseteq S^c$ with $|R| + |W| \leq s_0$ index exactly the elements $R \cup W$ of B_R , so $\bar{\pi}_\lambda(R) = \sum_W \pi_\lambda(R \cup W) \leq 2\pi_\lambda(R)$. The lower bound is $R \in B_R$. $\square$

Lemma 1.22 is the quantitative form of (P3): at every temperature, the d^w ways of adding w null covariates are beaten by the penalty $d^{-\kappa w}$, uniformly in λ , as soon as $\kappa > 1$. Everything in this subsection and the next rests on it.

1.5.3. Inside a block.

PROPOSITION 1.5. Assume 1.18 and 1.19. Then for every $\lambda \in [0, 1]$ and every $f : \mathcal{X} \rightarrow \mathbb{R}$,

$$(65) \quad \sum_{R \subseteq S} \bar{\pi}_\lambda(R) \text{Var}_{\pi_\lambda(\cdot | B_R)}(f) \leq 16 d s_0 \mathcal{E}_{K_\lambda, \pi_\lambda}(f),$$

that is, the first hypothesis of (17) holds with $\gamma_{\text{in}} = 1/(16ds_0)$.

PROOF. Fix $R \subseteq S$, write $\mu_R := \pi_\lambda(\cdot | B_R)$ and let K_R be the restriction of K_λ to B_R in the sense of (15).

Step 0: reduction. By Lemma 1.7ii and $K_\lambda(x, x) \geq 1/2$, Lemma 1.9 applies: each K_R is μ_R -reversible and non-negative definite, and it suffices to prove

$$(66) \quad \text{Gap}(K_R) \geq \frac{1}{16 d s_0} \quad \text{for every } R \subseteq S,$$

since the last claim of Lemma 1.9 then gives (65).

Step 1: the path ensemble. Identify B_R with $\{U \subseteq S^c : |U| \leq s_0 - |R|\}$ through $x = R \cup U$, and define

$$F(R \cup U) := R \cup U \setminus \{\max U\} \quad (U \neq \emptyset), \quad F(R) := R, \quad \Phi(R \cup U) := |U|.$$

Then $\Phi(F(x)) = \Phi(x) - 1$ when $\Phi(x) \geq 1$, and $\{\Phi = 0\}$ is the single state R , which is fixed by F ; so Definition 1.11 applies with $z_\bullet = R$. Every step of F is a single flip of a null coordinate, hence a proposal of (K2) and a fortiori an edge of K_R (both endpoints are in B_R). Let Γ^R be the ensemble generated by F . By Lemma 1.12,

$$(67) \quad \ell(\Gamma^R) \leq 2 \max_{x \in B_R} \Phi(x) \leq 2(s_0 - |R|) \leq 2s_0.$$

Step 2: edge weights. Let $z = R \cup V$ with $V \neq \emptyset$, let $j = \max V$ and let $e = (z, F(z))$ be the corresponding directed edge. By Lemma 1.22i applied to $x = R \cup V \setminus \{j\}$ and $W = \{j\}$ we have $\pi_\lambda(z) \leq \pi_\lambda(F(z))$, so $\min\{\pi_\lambda(z), \pi_\lambda(F(z))\} = \pi_\lambda(z)$ and Lemma 1.8 gives $\pi_\lambda(z) K_\lambda(z, F(z)) \geq \pi_\lambda(z)/(4d)$. Dividing by $\bar{\pi}_\lambda(R)$,

$$(68) \quad Q_R(e) := \mu_R(z) K_R(z, F(z)) \geq \frac{\mu_R(z)}{4d}.$$

Step 3: precedents. We claim that, with the convention $\max \emptyset = 0$,

$$(69) \quad \text{Prec}(z) = \{R \cup V \cup W : W \subseteq S^c \setminus V, \min W > \max V, |R| + |V| + |W| \leq s_0\}.$$

Indeed, the orbit of $R \cup U$ deletes the elements of U in decreasing order, so it passes through $R \cup V$ if and only if V is obtained from U by removing its largest elements, that is if and only if $U \supseteq V$ and every element of $U \setminus V$ exceeds every element of V . Consequently, by Lemma 1.22ii applied to $x = R \cup V$,

$$(70) \quad \frac{\mu_R[\text{Prec}(z)]}{\mu_R(z)} = \sum_W \frac{\pi_\lambda(R \cup V \cup W)}{\pi_\lambda(R \cup V)} \leq 2,$$

the sum over W in (69) being a subsum of that in (64). The same bound holds trivially for $z = R$, where $\text{Prec}(R) = B_R$ and $\mu_R[\text{Prec}(R)]/\mu_R(R) = 1/\mu_R(R) \leq 2$ by Lemma 1.22iii.

Step 4: conclusion. Combining Lemma 1.12, (68) and (70),

$$\varrho(\Gamma^R) \leq \max_{z \in B_R} \frac{\mu_R[\text{Prec}(z)]}{Q_R(z, F(z))} \leq \max_{z \in B_R} 4d \frac{\mu_R[\text{Prec}(z)]}{\mu_R(z)} \leq 8d.$$

With (67), Sinclair's bound (20) gives a Poincaré constant at least $(8d \cdot 2s_0)^{-1} = 1/(16ds_0)$ for (K_R, μ_R) , and K_R is non-negative definite, so this is (66). $\square$

Note where the two factors come from: the d is the cost of proposing a given flip among d possible ones, and the s_0 is the length of the longest path, which deletes at most s_0 null covariates and then adds at most s_0 . The exchange move of (K3) was never used: the only property of K_λ invoked is the single-flip edge bound (13). For the flip-only kernel, obtained by deleting (K3) and proposing a uniform flip with probability $1/2$, the proposal probability of a given flip is $1/(2d)$ rather than $1/(4d)$, so (13) holds a fortiori and Proposition 1.5 holds for that kernel with the same constant.

1.5.4. Between blocks. The projected chain lives on the 2^{s^*} subsets of S . We bound its Poincaré constant by the mass of its lightest state.

PROPOSITION 1.6. *Assume 1.18 and 1.19, and let $\bar{K}_\lambda$ be the projected kernel of Lemma 1.10 for the partition (62). Then the second hypothesis of (17) holds with*

$$(71) \quad \bar{\gamma} = \frac{\min_{R \subseteq S} \bar{\pi}_\lambda(R)}{8d s^*}.$$

PROOF. *Step 1: edge weights of the projected chain.* Let $R \subsetneq S$ and $j \in S \setminus R$, and set $R' = R \cup \{j\}$. By the definition of $\bar{K}_\lambda$,

$$\bar{Q}(R, R') := \bar{\pi}_\lambda(R) \bar{K}_\lambda(R, R') = \sum_{x \in B_R} \pi_\lambda(x) K_\lambda(x, B_{R'}) \geq \pi_\lambda(R) K_\lambda(R, R'),$$

since $R \in B_R$ and $R' \in B_{R'}$. The states R and R' differ by a single flip and $|R| < s^* \leq s_0$, so Lemma 1.8 and then Lemma 1.22iii give

$$(72) \quad \bar{Q}(R, R') \geq \frac{1}{4d} \min\{\pi_\lambda(R), \pi_\lambda(R')\} \geq \frac{1}{8d} \min\{\bar{\pi}_\lambda(R), \bar{\pi}_\lambda(R')\} \geq \frac{\min_{R'' \subseteq S} \bar{\pi}_\lambda(R'')}{8d}.$$

Step 2: paths. For $R \neq R'$ take the canonical path $R \rightarrow R \cap R' \rightarrow R'$, which first removes the elements of $R \setminus R'$ one at a time in increasing order and then adds those of $R' \setminus R$ one at a time in increasing order. All the states it visits are subsets of S , hence lie in $\mathcal{X}$; all the states it visits are distinct, because the first leg is strictly decreasing and the second strictly increasing for inclusion; and each of its edges is of the form treated in Step 1. Its length is $|R \setminus R'| + |R' \setminus R| \leq s^*$, so $\ell \leq s^*$.

Step 3: congestion. Bounding the sum over pairs by $\sum_{R,R'} \bar{\pi}_\lambda(R) \bar{\pi}_\lambda(R') = 1$ and using (72),

$$\varrho \leq \max_e \frac{1}{\bar{Q}(e)} \leq \frac{8d}{\min_R \bar{\pi}_\lambda(R)}.$$

Sinclair's bound (20) gives a Poincaré constant at least $\min_R \bar{\pi}_\lambda(R)/(8ds^*)$, which is (71). $\square$

LEMMA 1.23. Assume 1.18, 1.19 and (8), and place yourself on $\mathcal{H}_n$. Then for every $\lambda \in [0, 1]$,

$$(73) \quad \min_{R \subseteq S} \bar{\pi}_\lambda(R) \geq \frac{1}{2^{s^*+1}} d^{-\kappa s^*} e^{-\Omega_*}.$$

PROOF. Let $R, R' \subseteq S$. By Lemma 1.22iii, $\bar{\pi}_\lambda(R) \geq \pi_\lambda(R)$ and $\bar{\pi}_\lambda(R') \leq 2\pi_\lambda(R')$, so

$$(74) \quad \frac{\bar{\pi}_\lambda(R)}{\bar{\pi}_\lambda(R')} \geq \frac{1}{2} \frac{\pi_\lambda(R)}{\pi_\lambda(R')} = \frac{1}{2} d^{-\kappa(|R|-|R'|)} \left\{ \frac{m(R)}{m(R')} \right\}^\lambda \geq \frac{1}{2} d^{-\kappa s^*} \left\{ \frac{m(R)}{m(R')} \right\}^\lambda,$$

using $|R| - |R'| \leq s^*$. By Corollary 1.1 applied to the pair (R', R) of subsets of S , $\log\{m(R')/m(R)\} \leq \Omega_*$, hence $\{m(R)/m(R')\}^\lambda \geq e^{-\lambda\Omega_*} \geq e^{-\Omega_*}$ because $\Omega_* \geq 0$ and $\lambda \leq 1$. Therefore

$$(75) \quad \min_{R,R' \subseteq S} \frac{\bar{\pi}_\lambda(R)}{\bar{\pi}_\lambda(R')} \geq \frac{1}{2} d^{-\kappa s^*} e^{-\Omega_*}.$$

Finally, the 2^{s^*} numbers $\bar{\pi}_\lambda(R)$ sum to 1, so their maximum is at least 2^{-s^*} ; choosing R' to attain it in (75) gives $\min_R \bar{\pi}_\lambda(R) \geq 2^{-s^*} \cdot \frac{1}{2} d^{-\kappa s^*} e^{-\Omega_*}$. $\square$

1.5.5. The bound.

THEOREM 1.24. Assume 1.18, 1.19, and (8), and place yourself on the event $\mathcal{H}_n$, which has probability at least $1 - 6d^{-c}$. Then, with Ω_* as in (29), for **every** $\lambda \in [0, 1]$,

$$(76) \quad \text{Gap}(K_\lambda) \geq \frac{1}{1792 d^2 s_0 s^* 2^{s^*} d^{\kappa s^*} e^{\Omega_*}}.$$

In particular, if $\kappa, \alpha, L, \tilde{L}, r, c_2, s^* = \mathcal{O}(1)$ and the signal sits at the detection threshold in the sense of Remark 1.5, so that $e^{\Omega_*} = d^{\mathcal{O}(1)}$, then $\text{Gap}(K_\lambda) \geq d^{-\mathcal{O}(1)}/s_0$ uniformly in $\lambda \in [0, 1]$, and Assumption ?? holds along the whole schedule (48) with γ equal to the right-hand side of (76).

PROOF. Apply Lemma 1.10 with $\mu = \pi_\lambda$, $K = K_\lambda$ and the partition (62). Proposition 1.5 supplies the first hypothesis of (17) with $\gamma_{\text{in}} = 1/(16ds_0)$; Proposition 1.6 together with Lemma 1.23 supplies the second with

$$\bar{\gamma} = \frac{2^{-s^*-1} d^{-\kappa s^*} e^{-\Omega_*}}{8 d s^*}.$$

Both are at most 1, since $d \geq 1$, $s_0 \geq 1$ and $\Omega_* \geq 0$. Finally K_λ is non-negative definite by Lemma 1.7iv, so the last claim of Lemma 1.10 applies and

$$\text{Gap}(K_\lambda) \geq \frac{\gamma_{\text{in}} \bar{\gamma}}{7} = \frac{1}{7 \cdot 16 \cdot 8 \cdot 2} \cdot \frac{1}{d^2 s_0 s^* 2^{s^*} d^{\kappa s^*} e^{\Omega_*}},$$

and $7 \cdot 16 \cdot 8 \cdot 2 = 1792$. $\square$

REMARK 1.6 (What the bound costs). At $\lambda = 1$ the bound (76) is weaker than the bound $\mathcal{O}(1/(ds_0^2))$ of Theorem 2 of Yang, Wainwright and Jordan (2016) by a factor $d^{\mathcal{O}(s^*)}$. The two are not competing: theirs holds at one temperature and is sharp there, while (76) holds at every temperature, which is what Assumption ?? requires.

The loss has a single source, the bound of Lemma 1.23, which pays the full ratio between the heaviest and the lightest subset of S . That ratio is genuinely large near $\lambda = 1$, where $\bar{\pi}_1(\emptyset)$ is exponentially small in E ; what is not genuine is paying it, since a random walk on 2^{s^*} states whose profile is unimodal mixes in $\mathcal{O}(ds^*)$ steps regardless of how deep the well is. Making that precise requires a statement that the free energy $R \mapsto \log \bar{\pi}_\lambda(R)$ has no local maximum other than its mode, at every λ . That statement holds when the log-gains of the influential covariates are additive, that is when $\Delta_{\text{int}} = 0$; and it fails in the intermediate signal-to-noise example of Yang, Wainwright and Jordan (2016, Appendix B), where the chain is genuinely slowly mixing. So the factors 2^{s^*} and e^{Ω^*} in (76) cannot both be removed without a condition that rules out that example. Section 1.6 removes them under such a condition, by a different route, and Section 1.7 measures how much of what remains is real.

1.6. A temperature-adapted path ensemble. The bound of Theorem 1.24 pays, through Lemma 1.23, the full ratio between the heaviest and the lightest subset of S . We now remove that factor with a canonical path ensemble that moves with the temperature. The ensemble of Yang, Wainwright and Jordan (2016) is directed at x^* , which is the mode of π_λ only at $\lambda = 1$; ours is directed at the mode of π_λ itself.

1.6.1. Slices, moving centres and the margin condition. All the constructions of this subsection are carried out on a *slice* of $\mathcal{X}$ obtained by freezing some of the influential coordinates. This costs nothing when nothing is frozen, and it is what makes Theorem 1.32 an application of Proposition 1.7 rather than a repetition of its proof.

DEFINITION 1.25 (Slice and its centre). Let $C \subseteq S$ and $T \subseteq C$. The associated *slice* is

$$(77) \quad \mathcal{X}_T := \{x \in \mathcal{X} : x \cap C = T\},$$

and we write $\pi_{\lambda,T} := \pi_\lambda(\cdot \mid \mathcal{X}_T)$ and $K_{\lambda,T}$ for the restriction of K_λ to $\mathcal{X}_T$ in the sense of Lemma 1.9. The *centre* of the slice is

$$(78) \quad \hat{x}_{\lambda,T} := \arg \max \{ \pi_\lambda(v) : v \subseteq S, v \cap C = T \},$$

ties being broken by the lexicographic order. For $C = \emptyset$ we have $T = \emptyset$, $\mathcal{X}_\emptyset = \mathcal{X}$, $K_{\lambda,\emptyset} = K_\lambda$, and we write $\hat{x}_\lambda := \hat{x}_{\lambda,\emptyset}$ for the maximiser of π_λ over all subsets of S .

Note that $\hat{x}_{\lambda,T} \subseteq S$ and $\hat{x}_{\lambda,T} \cap C = T$, and that $\mathcal{X}_T$ is non-empty since it contains T . The defining property of the centre is the following pair of inequalities, obtained by comparing $\hat{x}_{\lambda,T}$ with its neighbours inside the competition set of (78).

LEMMA 1.26. For every $j \in S \setminus C$,

$$(79) \quad \lambda \psi_j(\hat{x}_{\lambda,T} \setminus \{j\}) \geq \kappa \log d \quad \text{if } j \in \hat{x}_{\lambda,T}, \quad \lambda \psi_j(\hat{x}_{\lambda,T}) \leq \kappa \log d \quad \text{if } j \notin \hat{x}_{\lambda,T}.$$

PROOF. Let $j \in S \setminus C$. If $j \in \hat{x}_{\lambda,T}$, then $v := \hat{x}_{\lambda,T} \setminus \{j\}$ is a subset of S with $v \cap C = \hat{x}_{\lambda,T} \cap C = T$, because $j \notin C$; so v is admissible in (78) and $\pi_\lambda(v) \leq \pi_\lambda(\hat{x}_{\lambda,T}) = \pi_\lambda(v \cup \{j\})$. By (31) this reads $\exp\{\lambda \psi_j(v) - \kappa \log d\} \geq 1$, which is the first inequality. If $j \notin \hat{x}_{\lambda,T}$, the same argument with $v := \hat{x}_{\lambda,T} \cup \{j\}$, which is again a subset of S meeting C in T , gives $\pi_\lambda(\hat{x}_{\lambda,T} \cup \{j\}) \leq \pi_\lambda(\hat{x}_{\lambda,T})$ and hence the second. $\square$

At $\lambda = 0$ and $C = \emptyset$ we have $\hat{x}_0 = \emptyset$, since $\pi_0 = p$ is strictly decreasing in $|x|$; at $\lambda = 1$, under Assumption 1.20, $\hat{x}_1 = x^*$, since $\psi_j^- > \kappa \log d$ makes the second inequality of (79) fail for every $j \notin \hat{x}_1$ unless $\hat{x}_1 = x^*$. In between, $\hat{x}_\lambda$ grows from $\emptyset$ to x^* , but not necessarily monotonically.

Recall the interaction spread Δ_{int} from (41). The ensemble below will move uphill at every step provided the temperature is not at a value where (79) is nearly an equality for some j . We formalise this as follows.

DEFINITION 1.27 (Margin condition). Say that $j \in S$ has margin at λ if

$$(M_\lambda) \quad \text{either } \lambda \psi_j^- \geq \kappa \log d + \log(2s^*) \quad \text{or} \quad \lambda \psi_j^+ \leq \kappa \log d - \log(2s^*).$$

Write $C_\lambda \subseteq S$ for the set of j that have no margin at λ , and $k_\lambda := |C_\lambda|$. We say that λ satisfies the margin condition if $k_\lambda = 0$.

In words: at temperature λ , covariate j is either worth including in every context, or worth including in no context, with a margin of $\log(2s^*)$ on the logarithmic scale. The role of that margin is to make a geometric series converge in Proposition 1.7; nothing else depends on its exact value.

1.6.2. The ensemble on a slice.

DEFINITION 1.28. In the setting of Definition 1.25, define $F_{\lambda,T} : \mathcal{X}_T \rightarrow \mathcal{X}_T$ by the first applicable rule, writing $\hat{x} := \hat{x}_{\lambda,T}$:

- (a) if $x \cap S^c \neq \emptyset$, delete $\max(x \cap S^c)$;
- (b) else if $\hat{x} \not\subseteq x$, add $\min(\hat{x} \setminus x)$;
- (c) else if $x \supsetneq \hat{x}$, delete $\min(x \setminus \hat{x})$;
- (d) else $F_{\lambda,T}(x) = x$.

We write $\Gamma^{\lambda,T}$ for the ensemble generated by $F_{\lambda,T}$ in the sense of Definition 1.11, and $F_\lambda := F_{\lambda,\emptyset}$, $\Gamma^\lambda := \Gamma^{\lambda,\emptyset}$ when $C = \emptyset$.

The order of the rules matters: null covariates are removed *first*, and only once none is left does the map adjust the influential ones. This is what makes the exchange move (K3) unnecessary, and it is also what makes the precedent count in Proposition 1.7 factorise.

LEMMA 1.29. $F_{\lambda,T}$ maps $\mathcal{X}_T$ into itself, and $\Gamma^{\lambda,T}$ is a canonical path ensemble for $(K_{\lambda,T}, \pi_{\lambda,T})$. Every edge it uses is a single-flip proposal of K_λ between two states of $\mathcal{X}_T$; no exchange proposal is needed; and $\ell(\Gamma^{\lambda,T}) \leq 2(s_0 + s^*) \leq 4s_0$. Moreover, rules (b) and (c) only ever flip a coordinate $j \in S \setminus C$.

PROOF. Write $\hat{x} = \hat{x}_{\lambda,T}$ and set

$$(80) \quad \Phi(x) := |x \cap S^c| + |\hat{x} \triangle (x \cap S)|,$$

where $\triangle$ is the symmetric difference. We verify the hypotheses of Definition 1.11 and Lemma 1.12.

The image lies in $\mathcal{X}_T$, and rules (b), (c) do not touch C . Rule (a) deletes an element of S^c , which leaves $x \cap C$ unchanged since $C \subseteq S$, and decreases $|x|$, so the image is in $\mathcal{X}_T$. Rules (b) and (c) fire only when $x \cap S^c = \emptyset$, that is $x \subseteq S$, so that $x \cap S = x$ and, x being in the slice, $x \cap C = T = \hat{x} \cap C$. Consequently

$$(81) \quad \hat{x} \setminus x \subseteq S \setminus C \quad \text{and} \quad x \setminus \hat{x} \subseteq S \setminus C,$$

because an element of C belongs to $\hat{x}$ if and only if it belongs to T , if and only if it belongs to x . So the index flipped by (b) or (c) lies in $S \setminus C$, the image still meets C in T , and it lies in $\mathcal{X}$: rule (c) deletes, while rule (b) produces a subset of S , of cardinality at most $s^* < s_0$.

Φ decreases by exactly one. Under rule (a) the first term of (80) drops by 1 and the second is unchanged, since only a coordinate of S^c moves. Under rules (b) and (c) the first term is 0 before and after, and the second drops by 1: rule (b) adds an element of $\hat{x} \setminus x$ and rule (c) removes an element of $x \setminus \hat{x}$, and in both cases the symmetric difference $\hat{x} \Delta x$ loses exactly that element.

The zero set. $\Phi(x) = 0$ forces $x \cap S^c = \emptyset$ and $x \cap S = \hat{x}$, that is $x = \hat{x}$; conversely $\Phi(\hat{x}) = 0$, and rule (d) fixes $\hat{x}$ and only $\hat{x}$, since if $x \subseteq S$ and $\hat{x} \subseteq x$ and $x \not\supseteq \hat{x}$ then $x = \hat{x}$.

Edges. Each rule changes exactly one coordinate, so each edge is a single flip; both of its endpoints lie in $\mathcal{X}_T$, so it is an edge of $K_{\lambda,T}$, and it carries positive weight by Lemma 1.8 and Lemma 1.7i.

Lemma 1.12 now applies and gives $\ell(\Gamma^{\lambda,T}) \leq 2 \max_x \Phi(x)$, and $\Phi(x) \leq |x \cap S^c| + s^* \leq s_0 + s^* \leq 2s_0$. $\square$

1.6.3. *The map never moves far downhill.* The next lemma is the reason the ensemble works at every temperature. It says that although $F_{\lambda,T}$ need not move uphill, it can never move far downhill, and that the bound involves the interaction spread Δ_{int} alone — not the ratio between the heaviest and the lightest subset of S , which is what Lemma 1.23 had to pay.

LEMMA 1.30. Assume 1.18 and 1.19, let $\lambda \in [0, 1]$, and let C, T be as in Definition 1.25. Let $z \in \mathcal{X}_T$ with $z \neq \hat{x}_{\lambda,T}$.

- (i) In all cases, $\pi_\lambda(z)/\pi_\lambda(F_{\lambda,T}(z)) \leq e^{\Delta_{\text{int}}}$.
- (ii) If z is treated by rule (a), then $\pi_\lambda(F_{\lambda,T}(z)) \geq d^{\kappa_\lambda} \pi_\lambda(z) \geq d^{1+c_\kappa} \pi_\lambda(z)$, with no margin condition on λ .
- (iii) If every $j \in S \setminus C$ has margin at λ and z is treated by rule (b) or (c), then $\pi_\lambda(F_{\lambda,T}(z)) \geq 2s^* \pi_\lambda(z)$.

PROOF. Write $\hat{x} = \hat{x}_{\lambda,T}$ and $F = F_{\lambda,T}$.

Rule (a). Here $j := \max(z \cap S^c) \in S^c$ and $F(z) = z \setminus \{j\}$. Applying (31) with $x = z \setminus \{j\}$, whose cardinality is $< s_0$,

$$\frac{\pi_\lambda(F(z))}{\pi_\lambda(z)} = \frac{\pi_\lambda(z \setminus \{j\})}{\pi_\lambda(z)} = \exp\{\kappa \log d - \lambda \psi_j(z \setminus \{j\})\} \geq d^{\kappa - \lambda(a_0 - \alpha)} = d^{\kappa_\lambda},$$

because $\psi_j(z \setminus \{j\}) \leq (a_0 - \alpha) \log d$ by Assumption 1.18 and $\lambda \geq 0$; and $d^{\kappa_\lambda} \geq d^{1+c_\kappa} > 1$ by Assumption 1.19. This proves ii, and also i in this case, since $d^{\kappa_\lambda} \geq 1 \geq e^{-\Delta_{\text{int}}}$.

Rule (b). Here $z \subseteq S$ and $j := \min(\hat{x} \setminus z)$, so that $j \in \hat{x}$ and, by (81), $j \in S \setminus C$. Both $\hat{x} \setminus \{j\}$ and z are subsets of $S \setminus \{j\}$, so (41) bounds ψ_j at either of them by $\psi_j^\pm$. By Lemma 1.26, $\lambda \psi_j(\hat{x} \setminus \{j\}) \geq \kappa \log d$, whence, using (31),

$$(82) \quad \log \frac{\pi_\lambda(z)}{\pi_\lambda(z \cup \{j\})} = \kappa \log d - \lambda \psi_j(z) \leq \lambda \{\psi_j(\hat{x} \setminus \{j\}) - \psi_j(z)\} \leq \lambda(\psi_j^+ - \psi_j^-) \leq \Delta_{\text{int}},$$

the last step because $\lambda \leq 1$ and $\psi_j^+ \geq \psi_j^-$.

Rule (c). Here $z \subseteq S$, $j := \min(z \setminus \hat{x})$, so $j \in S \setminus C$ by (81) and $j \notin \hat{x}$; both $z \setminus \{j\}$ and $\hat{x}$ are subsets of $S \setminus \{j\}$. By Lemma 1.26, $\lambda \psi_j(\hat{x}) \leq \kappa \log d$, whence

$$(83) \quad \log \frac{\pi_\lambda(z)}{\pi_\lambda(z \setminus \{j\})} = \lambda \psi_j(z \setminus \{j\}) - \kappa \log d \leq \lambda \{\psi_j(z \setminus \{j\}) - \psi_j(\hat{x})\} \leq \Delta_{\text{int}}.$$

This completes the proof of [i](#).

Proof of [iii](#). Suppose every $j \in S \setminus C$ has margin at λ . We claim first that for such j ,

$$(84) \quad j \in \hat{x} \implies \lambda\psi_j^- \geq \kappa \log d + \log(2s^*), \quad j \in S \setminus \hat{x} \implies \lambda\psi_j^+ \leq \kappa \log d - \log(2s^*).$$

Indeed, if $j \in \hat{x}$ then Lemma [1.26](#) and $\hat{x} \setminus \{j\} \subseteq S \setminus \{j\}$ give $\lambda\psi_j^+ \geq \lambda\psi_j(\hat{x} \setminus \{j\}) \geq \kappa \log d > \kappa \log d - \log(2s^*)$, since $2s^* \geq 2 > 1$; so the second alternative of (M_λ) fails and the first must hold. If $j \in S \setminus \hat{x}$ then, similarly, $\lambda\psi_j^- \leq \lambda\psi_j(\hat{x}) \leq \kappa \log d < \kappa \log d + \log(2s^*)$, so the first alternative fails and the second must hold.

Now revisit the two computations, noting that in both the flipped index j lies in $S \setminus C$ and therefore has margin. In rule [\(b\)](#) we have $j \in \hat{x}$, so by [\(84\)](#) the left-hand side of [\(82\)](#) is at most $\kappa \log d - \lambda\psi_j^- \leq -\log(2s^*)$. In rule [\(c\)](#) we have $j \notin \hat{x}$, so the left-hand side of [\(83\)](#) is at most $\lambda\psi_j^+ - \kappa \log d \leq -\log(2s^*)$. In both cases $\pi_\lambda(z)/\pi_\lambda(F(z)) \leq (2s^*)^{-1}$. $\square$

1.6.4. The bound on a slice.

PROPOSITION 1.7. *Assume [1.18](#) and [1.19](#). Let $\lambda \in [0, 1]$, let $C \subseteq S$ and $T \subseteq C$, and suppose that every $j \in S \setminus C$ has margin at λ in the sense of (M_λ) . Then*

$$(85) \quad \text{Gap}(K_{\lambda,T}) \geq \frac{1}{64 d s_0}.$$

PROOF. Write $\hat{x} = \hat{x}_{\lambda,T}$, $F = F_{\lambda,T}$, $\Gamma = \Gamma^{\lambda,T}$ and $\mu = \pi_{\lambda,T}$. We bound the congestion of Γ and apply Sinclair's bound [\(20\)](#), using Lemma [1.12](#) to reduce the computation to the directed edges $e = (z, F(z))$ with $z \neq \hat{x}$. Throughout, ratios $\mu(A)/\mu(B)$ equal $\pi_\lambda(A)/\pi_\lambda(B)$ for $A, B \subseteq \mathcal{X}_T$, since the normalising constant $\pi_\lambda(\mathcal{X}_T)$ cancels; we use the two interchangeably.

Step 1: edge weights. By hypothesis every $j \in S \setminus C$ has margin, so Lemma [1.30ii](#) and [iii](#) apply and give $\pi_\lambda(F(z)) \geq \pi_\lambda(z)$ for every $z \neq \hat{x}$, since $d^{\kappa_\lambda} \geq 1$ and $2s^* \geq 1$. The edge $(z, F(z))$ is a single flip between two states of $\mathcal{X}_T$ by Lemma [1.29](#), so Lemma [1.8](#), divided through by $\pi_\lambda(\mathcal{X}_T)$, gives

$$(86) \quad Q(e) := \mu(z)K_{\lambda,T}(z, F(z)) \geq \frac{1}{4d} \min\{\mu(z), \mu(F(z))\} = \frac{\mu(z)}{4d}.$$

Step 2: the structure of $\text{Prec}(z)$. Fix $z \in \mathcal{X}_T$ and recall from [\(21\)](#) that $\text{Prec}(z)$ is the set of $x \in \mathcal{X}_T$ whose orbit under F visits z . Because rule [\(a\)](#) takes precedence over the other two, the orbit of x consists of two phases: it first deletes the elements of $x \cap S^c$ in decreasing order, arriving at $x \cap S$ after $|x \cap S^c|$ steps, and thereafter moves among subsets of S under rules [\(b\)](#) and [\(c\)](#). Every state of the first phase meets S^c , except the last one; every state of the second phase is contained in S . We distinguish two cases accordingly, which are exhaustive.

Case A: $z \cap S^c \neq \emptyset$. Then z can occur only in the first phase of an orbit, and strictly before its last state. The first phase of the orbit of x passes through z if and only if z is obtained from x by deleting the largest elements of $x \cap S^c$, that is if and only if

$$(87) \quad x = z \cup W, \quad W \subseteq S^c \setminus z, \quad \min W > \max(z \cap S^c), \quad |z| + |W| \leq s_0.$$

Every such x lies in $\mathcal{X}_T$, because $W \subseteq S^c$ and $C \subseteq S$. Hence, by Lemma [1.22ii](#) applied at the state z , the sum over the W of [\(87\)](#) being a subsum of the one there,

$$(88) \quad \frac{\pi_\lambda[\text{Prec}(z)]}{\pi_\lambda(z)} = \sum_W \frac{\pi_\lambda(z \cup W)}{\pi_\lambda(z)} \leq 2.$$

Case B: $z \subseteq S$. Write $x = v \cup W$ with $v := x \cap S$ and $W := x \cap S^c$. The orbit of x reaches v at the end of its first phase and then follows the orbit of v ; so z lies on the orbit of x if and

only if $F^s(v) = z$ for some $s \geq 0$. Moreover $x \in \mathcal{X}_T$ if and only if $v \cap C = T$, again because $W \subseteq S^c$. Therefore

$$(89) \quad \frac{\pi_\lambda[\text{Prec}(z)]}{\pi_\lambda(z)} = \sum_{v \in \mathcal{V}} \frac{\pi_\lambda(v)}{\pi_\lambda(z)} \sum_W \frac{\pi_\lambda(v \cup W)}{\pi_\lambda(v)} \leq 2 \sum_{v \in \mathcal{V}} \frac{\pi_\lambda(v)}{\pi_\lambda(z)},$$

where $\mathcal{V} := \{v \subseteq S : v \cap C = T, F^s(v) = z \text{ for some } s \geq 0\}$, the inner sum runs over $W \subseteq S^c$ with $|v| + |W| \leq s_0$, and it was bounded by Lemma 1.22ii at the state v .

It remains to bound the outer sum, which we do by grouping the v according to the number s of steps needed to reach z ; write $\mathcal{V}_s := \{v \in \mathcal{V} : F^s(v) = z\}$, so that $\mathcal{V} = \bigcup_{s \geq 0} \mathcal{V}_s$, the union being disjoint because Φ strictly decreases along an orbit and hence $F^s(v) = F^{s'}(v) = z$ forces $s = s'$.

How many v are there in $\mathcal{V}_s$? Let $v \in \mathcal{V}_s$ and let $v = v_0, v_1, \dots, v_s = z$ be its orbit up to z . All the v_i are subsets of S , so all s steps use rule (b) or (c) and each flips one coordinate of $S \setminus C$, by Lemma 1.29. No coordinate is flipped twice: rule (b) fires while $\hat{x} \not\subseteq v_i$ and adds an element of $\hat{x}$, which no later application of rule (c) can remove, since rule (c) removes only elements of $v_i \setminus \hat{x}$; and once rule (c) has fired we have $\hat{x} \subseteq v_i$, a property preserved by further applications of rule (c), so rule (b) never fires again. Hence v and z differ in exactly s coordinates, all of them in $S \setminus C$, so v is determined by the set of those coordinates and

$$(90) \quad |\mathcal{V}_s| \leq \binom{|S \setminus C|}{s} \leq \binom{s^*}{s} \leq (s^*)^s.$$

How heavy are they? By Lemma 1.30iii, each of the s steps multiplies π_λ by at least $2s^*$, so $\pi_\lambda(z) = \pi_\lambda(F^s(v)) \geq (2s^*)^s \pi_\lambda(v)$. With (90) and $s^* \geq 1$,

$$(91) \quad \sum_{v \in \mathcal{V}} \frac{\pi_\lambda(v)}{\pi_\lambda(z)} = \sum_{s \geq 0} \sum_{v \in \mathcal{V}_s} \frac{\pi_\lambda(v)}{\pi_\lambda(z)} \leq \sum_{s \geq 0} (s^*)^s (2s^*)^{-s} = \sum_{s \geq 0} 2^{-s} = 2,$$

and (89) gives $\pi_\lambda[\text{Prec}(z)]/\pi_\lambda(z) \leq 4$. Since the bound of Case A is $2 \leq 4$, we have $\mu[\text{Prec}(z)]/\mu(z) \leq 4$ for every $z \in \mathcal{X}_T$.

Step 3: congestion and conclusion. By Lemma 1.12, (86) and Step 2,

$$\varrho(\Gamma) \leq \max_{z \in \mathcal{X}_T} \frac{\mu[\text{Prec}(z)]}{Q(z, F(z))} \leq \max_{z \in \mathcal{X}_T} 4d \frac{\mu[\text{Prec}(z)]}{\mu(z)} \leq 16d.$$

With $\ell(\Gamma) \leq 4s_0$ from Lemma 1.29, Sinclair's bound (20) gives a Poincaré constant at least $(16d \cdot 4s_0)^{-1}$ for $(K_{\lambda, T}, \mu)$, and $K_{\lambda, T}$ is non-negative definite by Lemma 1.9, so that Poincaré constant is $\text{Gap}(K_{\lambda, T})$. $\square$

THEOREM 1.31. *Assume 1.18 and 1.19, and let $\lambda \in [0, 1]$ satisfy the margin condition, that is $k_\lambda = 0$. Then*

$$(92) \quad \text{Gap}(K_\lambda) \geq \frac{1}{64d s_0}.$$

PROOF. Apply Proposition 1.7 with $C = \emptyset$ and $T = \emptyset$. Then $\mathcal{X}_\emptyset = \mathcal{X}$, $K_{\lambda, \emptyset} = K_\lambda$, and the hypothesis that every $j \in S \setminus C = S$ has margin at λ is exactly $k_\lambda = 0$. $\square$

REMARK 1.7 (Why this beats the bound at $\lambda = 1$). *Under Assumption 1.20 the margin condition holds at $\lambda = 1$, since $\psi_j^- \geq (\kappa + a_1) \log d = \kappa \log d + a_1 \log d \geq \kappa \log d + \log(2s^*)$ for every $j \in S$, by the requirement $d^{a_1} \geq 2s^*$; and it holds at $\lambda = 0$ as soon as $d^\kappa \geq 2s^*$, every j then taking the second alternative because $\lambda = 0$ makes its left-hand side vanish. So Theorem 1.31 applies at both endpoints of the path, and at $\lambda = 1$ it gives $\text{Gap}(K_1) \geq$*

$1/(64ds_0)$, a factor s_0 better than the $\mathcal{O}(1/(ds_0^2))$ of Theorem 2 of [Yang, Wainwright and Jordan \(2016\)](#).

The gain has one source. Their transition map replaces, at a saturated underfitted state, a null covariate by an influential one in a single exchange move, whose proposal probability is $1/(4ds_0)$ rather than $1/(4d)$ (Lemma 1.8). Ours never needs the exchange move, because rule (a) clears all null covariates before rule (b) adds any influential one, so no state on an orbit is ever saturated at the moment an influential covariate has to be added. The price is a longer path — $4s_0$ instead of $2s_0$ — which costs a factor 2, not a factor s_0 . A consequence is that Theorems 1.31 and 1.32, and also Proposition 1.5, hold for the flip-only kernel, which is cheaper and simpler than K_λ ; see item 5 of Section 1.10.

1.6.5. Temperatures at which the margin fails. The margin condition can fail, and it fails exactly where the mode $\hat{x}_\lambda$ is about to change. The next proposition locates those temperatures; the theorem after it covers them.

PROPOSITION 1.8. Assume 1.20, so that $\psi_j^- > 0$ for every $j \in S$, and assume $d^\kappa \geq 2s^*$. Then

$$\{\lambda \in [0, 1] : k_\lambda \geq 1\} \subseteq \bigcup_{j \in S} I_j, \quad I_j := \left(\frac{\kappa \log d - \log(2s^*)}{\psi_j^+}, \frac{\kappa \log d + \log(2s^*)}{\psi_j^-} \right),$$

a union of at most s^* open intervals, of lengths

$$|I_j| \leq \frac{\kappa \log d \Delta_{\text{int}}}{\psi_j^- \psi_j^+} + \frac{2 \log(2s^*)}{\psi_j^-}.$$

Neither $\lambda = 0$ nor $\lambda = 1$ belongs to this union.

PROOF. By Definition 1.27, $j \in C_\lambda$ means that both alternatives of (M_λ) fail, that is

$$\lambda \psi_j^- < \kappa \log d + \log(2s^*) \quad \text{and} \quad \lambda \psi_j^+ > \kappa \log d - \log(2s^*).$$

Since $0 < \psi_j^- \leq \psi_j^+$, these two inequalities are equivalent to $\lambda < (\kappa \log d + \log 2s^*)/\psi_j^-$ and $\lambda > (\kappa \log d - \log 2s^*)/\psi_j^+$, that is to $\lambda \in I_j$. For the length,

$$|I_j| = \frac{\kappa \log d + \log 2s^*}{\psi_j^-} - \frac{\kappa \log d - \log 2s^*}{\psi_j^+} = \kappa \log d \left(\frac{1}{\psi_j^-} - \frac{1}{\psi_j^+} \right) + \log(2s^*) \left(\frac{1}{\psi_j^-} + \frac{1}{\psi_j^+} \right),$$

and $1/\psi_j^- - 1/\psi_j^+ = (\psi_j^+ - \psi_j^-)/(\psi_j^- \psi_j^+) \leq \Delta_{\text{int}}/(\psi_j^- \psi_j^+)$, while $1/\psi_j^+ \leq 1/\psi_j^-$. Finally $0 \notin I_j$ because $\kappa \log d \geq \log(2s^*)$ makes the left endpoint non-negative, and $1 \notin I_j$ because Assumption 1.20 makes the right endpoint at most $(\kappa \log d + a_1 \log d)/\psi_j^- \leq 1$. $\square$

THEOREM 1.32. Assume 1.18 and 1.19. Then for **every** $\lambda \in [0, 1]$,

$$(93) \quad \text{Gap}(K_\lambda) \geq \frac{1}{7168 d^2 s_0 \max(k_\lambda, 1) (8s^* e^{\Delta_{\text{int}}})^{k_\lambda + 1}}.$$

PROOF. If $k_\lambda = 0$ this follows from Theorem 1.31, with room to spare, because $1/(64ds_0) \geq 1/(7168d^2s_08s^*e^{\Delta_{\text{int}}})$. Assume therefore $k_\lambda \geq 1$ and apply Lemma 1.10 to the partition of $\mathcal{X}$ indexed by $T \subseteq C_\lambda$,

$$(94) \quad B'_T := \{x \in \mathcal{X} : x \cap C_\lambda = T\}, \quad \bar{\pi}'_T(T) := \pi_\lambda(B'_T),$$

which has 2^{k_λ} blocks. Note that $B'_T = \bigsqcup \{B_R : R \subseteq S, R \cap C_\lambda = T\}$ in terms of the blocks (62), since $x \cap C_\lambda = (x \cap S) \cap C_\lambda$.

Step 1: inside a block. The blocks (94) are exactly the slices $\mathcal{X}_T$ of Definition 1.25 for the frozen set $C = C_\lambda$, and $K'_T := K_{\lambda,T}$ is the restriction of K_λ to $\mathcal{X}_T$ in the sense of (15). By the definition of C_λ , every $j \in S \setminus C_\lambda$ has margin at λ , so Proposition 1.7 applies with $C = C_\lambda$ and gives

$$\text{Gap}(K_{\lambda,T}) \geq \frac{1}{64 d s_0} \quad \text{for every } T \subseteq C_\lambda.$$

By the last claim of Lemma 1.9, the first hypothesis of (17) holds with $\gamma_{\text{in}} = 1/(64 d s_0)$.

Step 2: block mass ratios. Let $T \subseteq C_\lambda$ and $j \in C_\lambda \setminus T$, and set $T' = T \cup \{j\}$. The map $R \mapsto R \cup \{j\}$ is a bijection from $\mathcal{R}_T := \{R \subseteq S : R \cap C_\lambda = T\}$ onto $\mathcal{R}_{T'}$. For $R \in \mathcal{R}_T$ we have $R \subseteq S \setminus \{j\}$, so $\psi_j(R) \in [\psi_j^-, \psi_j^+]$, and since $j \in C_\lambda$ fails both alternatives of (M_λ) ,

$$\lambda \psi_j(R) - \kappa \log d \leq \lambda \psi_j^+ - \kappa \log d < \lambda \psi_j^+ - \lambda \psi_j^- + \log(2s^*) \leq \Delta_{\text{int}} + \log(2s^*),$$

and symmetrically $\lambda \psi_j(R) - \kappa \log d \geq \lambda \psi_j^- - \kappa \log d > -\Delta_{\text{int}} - \log(2s^*)$. By (31),

$$(95) \quad \frac{1}{2s^* e^{\Delta_{\text{int}}}} \leq \frac{\pi_\lambda(R \cup \{j\})}{\pi_\lambda(R)} \leq 2s^* e^{\Delta_{\text{int}}} \quad \text{for every } R \in \mathcal{R}_T.$$

Sum over $R \in \mathcal{R}_T$. If $a_R, b_R > 0$ satisfy $b_R/a_R \in [m, M]$ for every R , then $\sum_R b_R \leq M \sum_R a_R$ and $\sum_R b_R \geq m \sum_R a_R$, so $\sum_R b_R / \sum_R a_R \in [m, M]$; applying this with $a_R = \pi_\lambda(R)$ and $b_R = \pi_\lambda(R \cup \{j\})$, and using that $R \mapsto R \cup \{j\}$ is a bijection $\mathcal{R}_T \rightarrow \mathcal{R}_{T'}$,

$$(96) \quad \frac{1}{2s^* e^{\Delta_{\text{int}}}} \leq \frac{\sum_{R' \in \mathcal{R}_{T'}} \pi_\lambda(R')}{\sum_{R \in \mathcal{R}_T} \pi_\lambda(R)} \leq 2s^* e^{\Delta_{\text{int}}}.$$

By Lemma 1.22iii, summed over $R \in \mathcal{R}_T$,

$$(97) \quad \sum_{R \in \mathcal{R}_T} \pi_\lambda(R) \leq \bar{\pi}'_\lambda(T) \leq 2 \sum_{R \in \mathcal{R}_T} \pi_\lambda(R),$$

so (96) gives $\bar{\pi}'_\lambda(T')/\bar{\pi}'_\lambda(T) \in [(4s^* e^{\Delta_{\text{int}}})^{-1}, 4s^* e^{\Delta_{\text{int}}}]$. Any two subsets of C_λ are joined by at most k_λ such single flips, hence

$$(98) \quad \frac{\bar{\pi}'_\lambda(T)}{\bar{\pi}'_\lambda(T'')} \geq (4s^* e^{\Delta_{\text{int}}})^{-k_\lambda} \quad \text{for all } T, T'' \subseteq C_\lambda, \quad \text{so} \quad \min_T \bar{\pi}'_\lambda(T) \geq (8s^* e^{\Delta_{\text{int}}})^{-k_\lambda},$$

the last step because $\max_T \bar{\pi}'_\lambda(T) \geq 2^{-k_\lambda}$.

Step 3: between blocks. We bound the edge weights of the projected chain $\bar{K}'_\lambda$ on 2^{C_λ} . With T, j and T' as in Step 2, restricting the defining sum to the states $R \in \mathcal{R}_T$, all of which lie in B'_T and satisfy $R \cup \{j\} \in B'_{T'}$ with $|R| < s^* \leq s_0$,

$$\begin{aligned} \bar{Q}'(T, T') &= \sum_{x \in B'_T} \pi_\lambda(x) K_\lambda(x, B'_{T'}) \geq \sum_{R \in \mathcal{R}_T} \pi_\lambda(R) K_\lambda(R, R \cup \{j\}) \\ &\geq \frac{1}{4d} \sum_{R \in \mathcal{R}_T} \min\{\pi_\lambda(R), \pi_\lambda(R \cup \{j\})\} \geq \frac{1}{4d} \cdot \frac{1}{2s^* e^{\Delta_{\text{int}}}} \sum_{R \in \mathcal{R}_T} \pi_\lambda(R) \geq \frac{\bar{\pi}'_\lambda(T)}{16 d s^* e^{\Delta_{\text{int}}}}, \end{aligned}$$

using Lemma 1.8, then (95), then the upper bound in (97). Taking as canonical path from T to T'' the path $T \rightarrow T \cap T'' \rightarrow T''$, exactly as in Step 2 of the proof of Proposition 1.6, we get $\ell \leq k_\lambda$ and

$$\varrho \leq \max_T \frac{16 d s^* e^{\Delta_{\text{int}}}}{\bar{\pi}'_\lambda(T)} = \frac{16 d s^* e^{\Delta_{\text{int}}}}{\min_T \bar{\pi}'_\lambda(T)},$$

so that, by (20) and (98), the second hypothesis of (17) holds with

$$\bar{\gamma} = \frac{1}{16 d s^* e^{\Delta_{\text{int}}} k_{\lambda} (8 s^* e^{\Delta_{\text{int}}})^{k_{\lambda}}}.$$

Step 4: conclusion. Both γ_{in} and $\bar{\gamma}$ are at most 1, and K_{λ} is non-negative definite by Lemma 1.7iv, so Lemma 1.10 gives

$$\text{Gap}(K_{\lambda}) \geq \frac{\gamma_{\text{in}} \bar{\gamma}}{7} = \frac{1}{7 \cdot 64 \cdot 16 d^2 s_0 k_{\lambda} s^* e^{\Delta_{\text{int}}} (8 s^* e^{\Delta_{\text{int}}})^{k_{\lambda}}},$$

and $7 \cdot 64 \cdot 16 = 7168$. Finally $s^* e^{\Delta_{\text{int}}} \leq 8 s^* e^{\Delta_{\text{int}}}$, so the denominator is at most $7168 d^2 s_0 k_{\lambda} (8 s^* e^{\Delta_{\text{int}}})^{k_{\lambda}+1}$, which is (93). $\square$

REMARK 1.8 (Three bounds, and which to use). *At a given λ three bounds are now available and the best should be used.*

- Theorem 1.24 assumes only Assumption 1.18 but pays $d^{\mathcal{O}(s^*)}$.
- Theorem 1.31 gives $1/(64 d s_0)$, which is sharp, but needs (M_{λ}) ; by Proposition 1.8 this holds outside at most s^* intervals, and in particular at both endpoints of the path.
- Theorem 1.32 holds at every λ and pays $(8 s^* e^{\Delta_{\text{int}}})^{k_{\lambda}+1}$, a quantity that does not grow with d once $\Delta_{\text{int}} = \mathcal{O}(1)$ and $k_{\lambda} = \mathcal{O}(1)$.

The third is the useful one: it removes the $d^{\mathcal{O}(s^)}$ of Theorem 1.24 entirely, at the cost of the interaction spread Δ_{int} , which is a property of the design alone and vanishes when the residualised influential columns $(I - P_{S \setminus \{j\}})X_j$, $j \in S$, are orthogonal. On the instances of Section 1.7 the factor $(8 s^* e^{\Delta_{\text{int}}})^{k_{\lambda}+1}$ exceeds 10^{30} while the exact gap varies by less than a factor two along the whole path, so there at least it is an artefact of the proof. Since Assumption ?? is applied with $\gamma = \min_t \text{Gap}(K_{\lambda_t})$, what matters is the worst λ along the schedule, and hence $\max_{\lambda} k_{\lambda}$; nothing prevents a practitioner from choosing the schedule so as to avoid the intervals of Proposition 1.8, although one would then have to estimate $\psi_j^{\pm}$.*

1.7. Numerical illustration. The instances of this subsection are small enough that everything can be computed exactly: the state space $\mathcal{X}$ is enumerated, the targets π_{λ} are evaluated at every state, the kernel K_{λ} of Section 1.1.4 is assembled, and its spectral gap is obtained by diagonalisation. No Monte Carlo is involved, so what is plotted is $\text{Gap}(K_{\lambda})$ itself and not an estimate of it. The code is `binary_numerics.py`.

1.7.0.1. *Setting.* We take $n = 100$, $d = 14$, $s_0 = 4$, so that $|\mathcal{X}| = \sum_{k \leq 4} \binom{14}{k} = 1471$; $S = \{1, 2, 3\}$, $s^* = 3$; $\sigma_0 = 1$; $\kappa = 2$ and $g = d^{2\alpha}$ with $\alpha = 3/2$. The influential columns are equicorrelated with correlation 0.3, which makes the interaction spread Δ_{int} of (41) non-zero; the null columns are residualised against $\text{span}(X_S)$, so the proxy energy Ξ of (39) is exactly 0 and Proposition 1.2 applies in its favourable form. All columns are normalised to $\|X_j\|_2^2 = n$. We set $\beta_j^* = C_{\beta}$ for $j \in S$ and $\beta_j^* = 0$ otherwise, and sweep C_{β} through the range in which the posterior mass $\pi(x^* | y)$ moves from 0 to 1.

The assumptions can be checked rather than assumed. The smallest a_0 for which Assumption 1.18 holds, namely $a_0 = \alpha + \max \psi_j(x) / \log d$ over $x \in \mathcal{X}$ with $|x| < s_0$ and $j \in S^c \setminus x$, stays in $[0.506, 0.538]$ over the whole sweep, well below $\alpha = 3/2$; the sparsity budget (37) therefore holds with slack $\min(\kappa, \kappa + \alpha - a_0) - 1 = 1$, the binding constraint being the one at $\lambda = 0$, and $\lambda_c = 1$ in (38). As a check on the eigensolver, the gap at $\lambda = 1$ computed by the sparse deflated solver and by dense diagonalisation agree to seven significant figures.

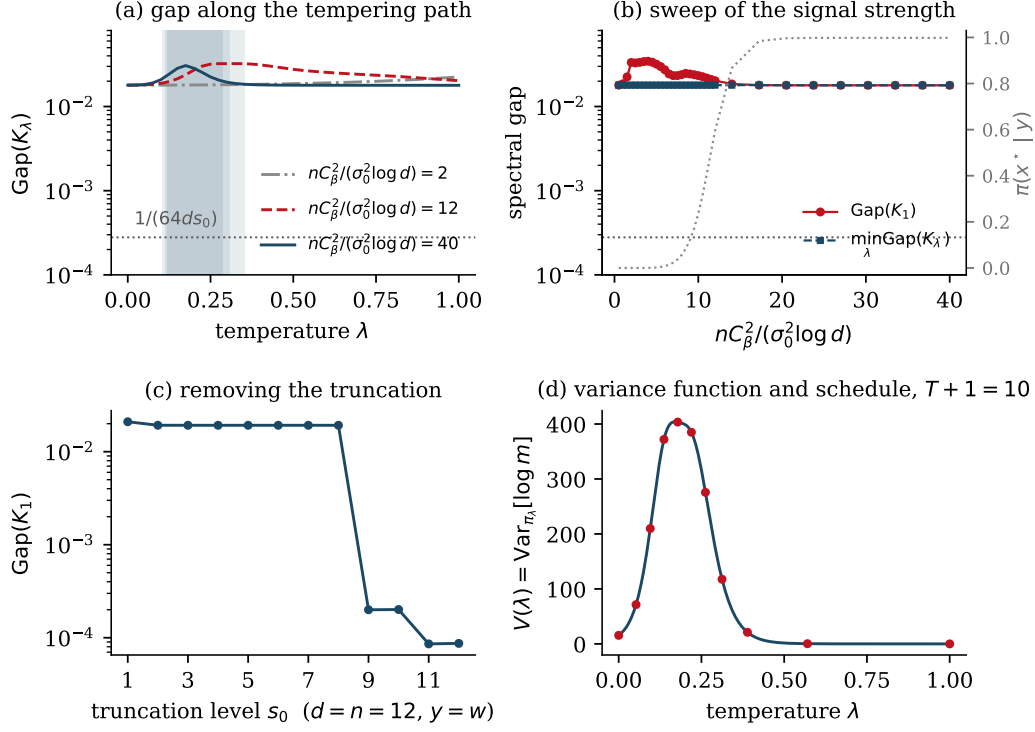


Fig 1: Exact computations on small instances. **(a)** $\text{Gap}(K_\lambda)$ along the tempering path (10) for three signal strengths; the shaded band is the union of the intervals of Proposition 1.8 on which the margin condition (M_λ) fails, for the strongest signal; the dotted line is the bound $1/(64ds_0)$ of Theorem 1.31. **(b)** $\text{Gap}(K_1)$ and $\min_{\lambda \in [0,1]} \text{Gap}(K_\lambda)$ as the signal strength sweeps through the range in which the posterior mass $\pi(x^* | y)$ (dotted, right axis) moves from 0 to 1. **(c)** $\text{Gap}(K_1)$ as a function of the truncation level s_0 for the example of Yang, Wainwright and Jordan (2016, Appendix B): $d = n = 12$ and $y = w$ pure noise. **(d)** the variance function V of (46) and the schedule (48) it produces (dots).

1.7.0.2. (a) The gap along the path. Over the whole sweep and the whole path, $\text{Gap}(K_\lambda)$ stays in $[1.79 \times 10^{-2}, 3.45 \times 10^{-2}]$: it varies by less than a factor two, in λ and in the signal strength together. The bound $1/(64ds_0) = 2.79 \times 10^{-4}$ of Theorem 1.31 holds, with a slack between 64 and 124. The slack is accounted for by the two crude steps of that proof: at $\lambda = 0$ the measured gap is 1.7948×10^{-2} , against $1/(4d) = 1.7857 \times 10^{-2}$, so the chain essentially realises the single-flip proposal probability of Lemma 1.8, and what the bound gives away is the congestion factor $\varrho \leq 16d$ and the length factor $\ell \leq 4s_0$ of Theorem 1.31, whose product is $64s_0$.

The shaded band is where the margin condition (M_λ) fails. For the strongest signal it is the union of the three intervals $(0.114, 0.353)$, $(0.119, 0.286)$, $(0.104, 0.310)$, which Proposition 1.8 predicts to be contained in $[0, 1]$ and to exclude both endpoints, and on part of which all three influential covariates fail simultaneously, so $k_\lambda = 3$. With $\Delta_{\text{int}} = 10.6$ there, the factor $(8s^*e^{\Delta_{\text{int}}})^{k_\lambda+1}$ of Theorem 1.32 exceeds 10^{30} . The measured gap on that band is not smaller than elsewhere; it is slightly larger. On this instance the price paid by Theorem 1.32 is therefore an artefact of the proof and not a feature of the chain. That is a statement about this instance only; we do not claim that $e^{\Delta_{\text{int}}}$ can be removed in general, and Remark 1.6 recalls why some such factor must remain.

1.7.0.3. (b) *Sweeping the signal strength.* As $nC_\beta^2/(\sigma_0^2 \log d)$ runs from 0.5 to 40, the posterior mass $\pi(x^* | y)$ moves from 0 to 0.998, so the sweep does cross the range in which the model is neither clearly identified nor clearly unidentifiable. Neither $\text{Gap}(K_1)$ nor $\min_\lambda \text{Gap}(K_\lambda)$ moves by more than a factor two across the sweep, and both stay two orders of magnitude above the bound.

Two caveats. First, $\min_\lambda \text{Gap}(K_\lambda)$ is attained at $\lambda = 0$ for most of the sweep, and $\pi_0 = p$ does not depend on the data, so the flatness of that curve is in part by construction; the informative curve is $\text{Gap}(K_1)$. Second, this is one design at one small dimension, and a gap that does not collapse here is not evidence that it cannot collapse. Panel (c) exhibits an instance where $\text{Gap}(K_1)$ does collapse. Nor does anything here compare SMC with MCMC: the cost of SMC involves the horizon T and, for standard SMC, the overhead $\mathcal{D}$ of Lemma 1.33, none of which is a spectral gap.

1.7.0.4. (c) *Removing the truncation.* Yang, Wainwright and Jordan (2016, Appendix B) show that with $d = n$, with $y = w$ pure noise, and with the sparsity prior left untruncated, the saturated model becomes a trap: it has $R^2 = 1$, so its marginal likelihood is large, and every neighbour is exponentially lighter. Panel (c) reproduces this with $d = n = 12$, $\kappa = 2$, $\alpha = 3/2$, sweeping the truncation level. For $s_0 \leq 8$ the gap is 1.92×10^{-2} , independent of s_0 ; at $s_0 = 9$ it drops to 2.0×10^{-4} and at $s_0 = d = 12$ to 8.7×10^{-5} , a factor 221. This is the concrete content of item 2 of Section 1.10: the constraint $|x| \leq s_0$ in the prior (3) is not a technical convenience, and the uniform prior on $\{0, 1\}^d$ used by Schäfer and Chopin (2013) sits on the wrong side of it.

1.7.0.5. (d) *The schedule.* The variance function V of (46) is concentrated near $\lambda \approx 0.2$, which is where the mass of π_λ leaves the null model, and it is nearly zero on $[0.5, 1]$. The schedule (48) places its steps accordingly: ten steps in all, of which eight lie in $[0, 0.4]$, the last one covering $[0.57, 1]$. The crude bound (49) gives $\lceil \sqrt{V/\log 2} \rceil = 25$, so it is loose by a factor 2.5 here, which is what one expects from a bound that pays the worst variance along the whole path.

1.8. *From spectral gaps to β -mixing times.* Theorems ??, ?? and ??, which govern standard SMC, are stated in terms of the β -mixing time τ_β rather than of the spectral gap. Section ?? showed that the two can be far apart in either direction. On a finite space they differ by at most $\frac{1}{2} \log |\mathcal{X}|$, and in the direction unfavourable to standard SMC.

LEMMA 1.33. *Let $\mathcal{X}$ be finite and let K be reversible and non-negative definite with respect to a probability measure $\pi_\bullet$ of full support, with spectral gap $\gamma_\bullet > 0$. Then for all $\xi \in (0, 1)$ and $\omega \geq 1$,*

$$(99) \quad \tau_\beta(\xi, \omega, K) \leq \frac{1}{\gamma_\bullet} \left(\frac{1}{2} \log |\mathcal{X}| + \log \frac{\omega}{2\xi} \right).$$

PROOF. *Step 1: from χ^2 to total variation.* For any probability measure $\mu \ll \pi_\bullet$, Cauchy–Schwarz gives

$$2\text{TV}(\mu, \pi_\bullet) = \sum_z \pi_\bullet(z) \left| \frac{\mu(z)}{\pi_\bullet(z)} - 1 \right| \leq \left\{ \sum_z \pi_\bullet(z) \left(\frac{\mu(z)}{\pi_\bullet(z)} - 1 \right)^2 \right\}^{1/2} = \chi^2(\mu | \pi_\bullet)^{1/2}.$$

Step 2: contraction. A $\pi_\bullet$ -reversible, non-negative definite kernel with spectral gap $\gamma_\bullet$ has operator norm $1 - \gamma_\bullet$ on $\mathcal{L}_{0, \pi_\bullet}^2$; applying this to $h = d\mu/d\pi_\bullet - 1$, whose $\mathcal{L}^2$ -norm squared is $\chi^2(\mu | \pi_\bullet)$, gives $\chi^2(\mu K | \pi_\bullet) \leq (1 - \gamma_\bullet)^2 \chi^2(\mu | \pi_\bullet)$ and, by iteration,

$$\chi^2(\delta_x K^q | \pi_\bullet) \leq (1 - \gamma_\bullet)^{2q} \chi^2(\delta_x | \pi_\bullet) = (1 - \gamma_\bullet)^{2q} \left\{ \frac{1}{\pi_\bullet(x)} - 1 \right\}.$$

With Step 1, $\text{TV}(\delta_x K^q, \pi_\bullet) \leq \frac{1}{2}(1 - \gamma_\bullet)^q \pi_\bullet(x)^{-1/2}$.

Step 3: averaging over a warm start. Let μ be ω -warm, that is $\mu(x) \leq \omega \pi_\bullet(x)$ for every x .

Then

$$\int \text{TV}(\delta_x K^q, \pi_\bullet) \mu(dx) \leq \frac{\omega}{2}(1 - \gamma_\bullet)^q \sum_{x \in \mathcal{X}} \pi_\bullet(x) \pi_\bullet(x)^{-1/2} = \frac{\omega}{2}(1 - \gamma_\bullet)^q \sum_x \pi_\bullet(x)^{1/2} \leq \frac{\omega}{2}(1 - \gamma_\bullet)^q |\mathcal{X}|^{1/2},$$

the last step by Cauchy–Schwarz, $\sum_x \pi_\bullet(x)^{1/2} \leq \{|\mathcal{X}| \sum_x \pi_\bullet(x)\}^{1/2}$. Using $(1 - \gamma_\bullet)^q \leq e^{-\gamma_\bullet q}$ and requiring the right-hand side to be at most ξ gives (99). $\square$

Since $|\mathcal{X}| = \sum_{k \leq s_0} \binom{d}{k} \leq (ed/s_0)^{s_0}$, the overhead of τ_β over $1/\gamma_\bullet$ is at most

$$(100) \quad \mathcal{D} := \frac{1}{2} \log |\mathcal{X}| \leq \frac{s_0}{2} \log \frac{ed}{s_0}.$$

It is a power of s_0 , not a logarithm of it, and it is the price standard SMC pays over waste-free SMC on this problem. The contrast with the continuous case of Section ?? is instructive: there, a warm-start device (Proposition ??) made τ_β smaller than γ^{-1} ; here $\mathcal{D}$ comes from $\min_x \pi_\bullet(x)$ being exponentially small in s_0 , and no analogue of that device is available.

1.9. Complexity. We now collect the three inputs. Write

$$(101) \quad \mathcal{G} := \min \left\{ 1792 d^2 s_0 s^* 2^{s^*} d^{\kappa s^*} e^{\Omega_*}, 7168 d^2 s_0 \max(k_*, 1) (8s^* e^{\Delta_{\text{int}}})^{k_*+1} \right\}, \quad k_* := \max_{t=0, \dots, T} k_{\lambda_t},$$

for the inverse spectral gap, the better of Theorems 1.24 and 1.32 along the schedule; $\mathcal{D}$ as in (100) for the mixing-time overhead of Lemma 1.33; and $T \leq C_V \sqrt{\mathcal{J}_* / \log 2}$ for the horizon of Proposition 1.4iii with Corollary 1.2.

THEOREM 1.34. *Assume the hypotheses of Theorem 1.24 and of Corollary 1.2, and run the samplers of ???????? on the path (10) with the schedule (48) and the kernels K_{λ_t} . Then, on the event $\mathcal{H}_n$, Assumptions ?? and ?? hold with $\gamma \geq 1/\mathcal{G}$ as in (101), and the number of Markov steps sufficient for (??), respectively for (??), is bounded as follows, up to factors logarithmic in T , $1/\eta$ and $1/\varepsilon$.*

<i>Estimate</i>	<i>Algorithm</i>	<i>Markov steps</i>
$\widehat{\pi}_{T-1}(f)$	Alg. ??	$T \mathcal{G} \mathcal{D} \varepsilon^{-2}$
	Alg. ??	$T \mathcal{G} \varepsilon^{-2}$
	Alg. ??	$\mathcal{G} (T + \varepsilon^{-2})$
$\widehat{Z}_T$	Alg. ??	$T^4 \mathcal{G} \mathcal{D} \varepsilon^{-2}$
	Alg. ?? with medians	$T^3 \mathcal{G} \mathcal{D} \varepsilon^{-2}$
	Alg. ??	$T^4 \mathcal{G} \varepsilon^{-2}$
	Alg. ??	$T^3 \mathcal{G} \varepsilon^{-2}$

The rows for $\hat{Z}_T$ using Algorithms ?? and ?? hold with $\eta = 1/4$.

PROOF. Assumption ?? holds along the schedule (48) by Proposition 1.4ii, and Assumption ?? holds with $\gamma \geq 1/\mathcal{G}$ by Theorems 1.24 and 1.32, applied at each $\lambda_t \in [0, 1]$. Since K_{λ_t} is non-negative definite, Lemma 1.33 applies and gives, for every t ,

$$(102) \quad \tau_\beta(\xi, 2, K_{\lambda_t}) \leq \mathcal{G}\{\mathcal{D} + \log(1/\xi)\},$$

using $\log(2/(2\xi)) = \log(1/\xi)$; hence the same bound for the uniform $\tau_\beta(\xi, 2)$ of Section ?? . The total number of Markov steps is $(T+1)MP$ for Algorithms ?? and ??, $M \sum_t P_t$ for Algorithm ??, and $J(T+1)MP$ for Algorithm ?? . The seven rows follow.

- Row 1. Theorem ?? with $\bar{\chi}^2 = 2$ needs $M = \mathcal{O}(\varepsilon^{-2} \log(T/\eta))$ and $P \geq \tau_\beta(\eta/(2MT), 2) = \mathcal{O}(\mathcal{G}\{\mathcal{D} + \log(MT/\eta)\})$ by (102). Hence $(T+1)MP = \tilde{\mathcal{O}}(T\mathcal{G}\mathcal{D}\varepsilon^{-2})$.
- Rows 2 and 3. Theorems ?? and ?? with $M = 1$ and $\gamma^{-1} \leq \mathcal{G}$; for Row 3 the greedy schedule has $P_t = \mathcal{O}(\mathcal{G} \log(T/\eta))$ for $t < T$ and $P_T = \mathcal{O}(\mathcal{G}\varepsilon^{-2} \log(T/\eta))$, whose sum is $\tilde{\mathcal{O}}(\mathcal{G}(T + \varepsilon^{-2}))$.
- Row 4. Theorem ?? with $M = \Omega(T^3\varepsilon^{-2})$ and $P = \Omega(\tau_\beta(1/(MT)))$.
- Row 5. Theorem ?? with $J = \Omega(\log(T/\eta))$, $M = \Omega(T^2\varepsilon^{-2})$ and $P = \Omega(\tau_\beta(1/(MT)))$.
- Rows 6 and 7. Theorems ?? and ?? with $M = 1$ and, respectively, $P = \Omega(T^3\gamma^{-1}\varepsilon^{-2})$ and $P = \Omega(T^2\gamma^{-1}\varepsilon^{-2})$ with $J = \Omega(\log(T/\eta))$. $\square$

COROLLARY 1.3 (The polynomial regime). *Take $\kappa, \alpha, L, \tilde{L}, C_m, r, c_2, s^*, C_V, \Delta_{\text{int}}, k_\star = \mathcal{O}(1)$ and a signal at the detection threshold in the sense of Remark 1.5, so that $\mathbf{E} = \mathcal{O}(\log d)$, $\Omega_\star = \mathcal{J}_\star = \mathcal{O}(\log d)$, $\mathcal{G} = \mathcal{O}(s_0 d^q)$ for a constant q , $\mathcal{D} = \mathcal{O}(s_0 \log d)$ and $T = \mathcal{O}(\sqrt{\log d})$. Then the marginal likelihood is estimated within relative error ε with probability $1 - \eta$ using $\tilde{\mathcal{O}}(d^q s_0 (\log d)^{3/2} \varepsilon^{-2})$ Markov steps by Algorithm ??, that is $\tilde{\mathcal{O}}(n d^q s_0^2 (\log d)^{3/2} \varepsilon^{-2})$ operations, and $\pi(f)$ is estimated within absolute error ε using $\tilde{\mathcal{O}}(d^q s_0 \{\sqrt{\log d} + \varepsilon^{-2}\})$ Markov steps by Algorithm ?? . Both are polynomial in $(d, n, s_0, \varepsilon^{-1})$.*

PROOF. Substitute the stated orders into the table of Theorem 1.34, using $T^3 = \mathcal{O}((\log d)^{3/2})$ for Algorithm ?? and $T = \mathcal{O}(\sqrt{\log d})$ for Algorithm ?? . The conversion into operations multiplies by the $\mathcal{O}(ns_0 + s_0^2) = \mathcal{O}(ns_0)$ cost of one evaluation of m , valid since $s_0 < n$. $\square$

1.9.0.1. *Comparison with MCMC.* Corollary 1 of Yang, Wainwright and Jordan (2016) states that the chain K_1 started at the worst state returns $x^\star$ after $\mathcal{O}(\alpha n d s_0^2 \log d)$ steps. Consider instead starting it at ν , which is the only practical option and the one SMC uses. By the contraction inequality of Step 2 in the proof of Lemma 1.33, the number of steps needed for the law of the chain to be within ξ of π is of order $\gamma^{-1}\{\ell_\nu + \log(1/\xi)\}$ with $\ell_\nu := \log \chi^2(\nu | \pi)$, and estimating $\pi(f)$ to accuracy ε costs $\mathcal{O}(\gamma^{-1}\varepsilon^{-2})$ further steps. Now $\chi^2(\nu | \pi) = \mathbb{E}_\nu[d\nu/d\pi]$, so Jensen's inequality gives

$$\ell_\nu \geq \mathbb{E}_\nu \left[\log \frac{d\nu}{d\pi} \right] = \text{KL}(\nu | \pi) \geq \mathcal{J} - \text{KL}(\pi | \nu).$$

So MCMC pays $\tilde{\mathcal{O}}(\mathcal{G}\{\ell_\nu + \varepsilon^{-2}\})$ where Algorithm ?? pays $\tilde{\mathcal{O}}(\mathcal{G}\{\sqrt{\mathcal{J}} + \varepsilon^{-2}\})$: the cost of reaching the target from the prior falls from an amount proportional to $\mathcal{J}$ to one proportional to $\sqrt{\mathcal{J}}$. This is the general phenomenon of Section ??, here made explicit. Moreover MCMC returns no estimate of Z_T with a comparable guarantee, which is the reason Schäfer and Chopin (2013) use SMC on this problem in the first place.

1.10. Consequences for practice.

1. *Start from the prior, and temper the marginal likelihood only.* This is the path (10). Property (P3) is what makes Theorems 1.24 and 1.32 hold at every temperature: the penalty $d^{-\kappa}$ charged for each null covariate is the same at every λ , so the d ways of adding one are beaten uniformly in λ as soon as $\kappa > 1$ (Lemma 1.22). Tempering the whole posterior, $\pi_\lambda \propto (pm)^\lambda$, destroys this. The initial distribution is then sampled exactly by (P1), and $\hat{Z}_T$ estimates the marginal likelihood directly.
2. *Take $\kappa > 1$ and truncate at $s_0 = \mathcal{O}(n/\log d)$.* Schäfer and Chopin (2013) take the uniform prior on $\{0, 1\}^d$, that is $\kappa = 0$ and $s_0 = d$. Then the $\lambda = 0$ half of the sparsity budget (37) fails, Lemma 1.22 fails with it, and so does every bound of Sections 1.5 and 1.6; Assumption 1.5 fails as well. Both are repaired by the prior (3) with $\kappa > 1$ and s_0 of the order that Assumption 1.5 allows. This is not only a limitation of the proof: panel (c) of Figure 1 shows the exact gap of the local kernel dropping by a factor 221 when the truncation is removed, on the example of Yang, Wainwright and Jordan (2016, Appendix B).
3. *Spend the sparsity budget: take $\kappa + \alpha \geq 1 + a_0 + c_\kappa$.* This is Assumption 1.19 at $\lambda = 1$, in the form (37): the sparsity penalty $d^{-\kappa}$ and the Occam factor $(1 + g)^{-|x|/2} = d^{-\alpha + \mathcal{O}(1)}$ must together beat both the d ways of adding a covariate and the largest amount of likelihood d^{a_0} that a null covariate can buy. By Propositions 1.1 and 1.2, $\kappa + \alpha \gtrsim L + \tilde{L}$ suffices when the proxy energy Ξ of (39) is $\mathcal{O}(\log d)$, and $\kappa + \alpha = \Omega(s^*)$ is needed in the worst case. Since Assumption 1.4 constrains only $\kappa + \alpha$ from below as well, and since κ is also constrained on its own by the $\lambda = 0$ half of (37), the practical recipe is: fix κ slightly above 1 for the prior end of the path, and then raise α , that is g , until the budget holds at $\lambda = 1$. Raising α is free.
4. *The local kernel has a polynomial but large inverse gap.* Theorem 1.24 gives $\gamma^{-1} \leq \mathcal{G}$, which without the margin condition is $d^{2+\kappa s^*}$ up to factors. This is polynomial, and it is also large enough to explain the observation of Schäfer and Chopin (2013) that their sampler does not work with local kernels. Their global kernel, an independent Metropolis–Hastings kernel with a fitted proposal, has spectral gap equal to $\min_x q_{\theta_t}(x)/\pi_{t-1}(x)$, for which we have no lower bound and which can be exponentially small in d ; we prove nothing about it.
5. *The exchange move is not needed.* The ensemble of Definition 1.28 uses single flips only, because it deletes null covariates before it adjusts the influential ones. So Theorems 1.31 and 1.32, and also Proposition 1.5, hold for the flip-only kernel, which is cheaper per step and simpler to implement than K_λ . As explained in Remark 1.7, the exchange move is what costs Yang, Wainwright and Jordan (2016) the second factor s_0 in their $\mathcal{O}(1/(ds_0^2))$.
6. *Use Algorithm ?? for moments and Algorithm ?? for the marginal likelihood.* This is the recommendation of Section ??, and it is reinforced here: standard SMC has no compensating advantage on this problem, because the factor $\mathcal{D} = \mathcal{O}(s_0 \log d)$ of Lemma 1.33 comes from $\min_x \pi_\bullet(x)$ being exponentially small in s_0 and no analogue of Proposition ?? removes it.

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